Framework for Blockchain-Based Community-Assisted Privacy- Preserving Reporting Service for Local Governance

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Abstract

Local governments in developing countries often struggle to address public issues efficiently. These struggles are mainly due to limited transparency, centralized data management, and insufficient citizen participation. E-governance and crowd-sourced reporting systems are designed to enhance public engagement. But their effectiveness is limited by issues related to data integrity, user anonymity, content reliability, and trustworthiness.As a solution to the above challenges, we propose a blockchain-based, community-assisted, privacy-preserving reporting framework for local governance. The main objective of the project is to design and prototype a permissioned blockchain-based reporting system. This system will enable citizens to anonymously submit public issue reports and participate in opinion polling. Also, our system will ensure data immutability and transparency. To manage reporting workflows, issue tracking, and community voting in a secure and tamper-resistant manner, the blockchain will use Proof-of-Authority (PoA) mechanism and smart contracts. User identity is decoupled from the report content using a simulated identity and an access control mechanism to preserve the privacy of the users. Large data objects, such as images, are stored off-chain with cryptographic hashes anchored on the blockchain. Additionally, an Artificial Intelligence (AI)-assisted content moderation mechanism will be used to minimize the impact of malicious, low-quality reporting content. To demonstrate the feasibility of the proposed framework, A web-based Decentralized App (DApp) is developed as a proof of concept within a controlled experimental environment. Through this project, we aim to deliver a practical and scalable prototype that enhances citizen participation, accountability, and trust in local governance systems.

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Contents

Abstract i

Acknowledgements ii

Contents vi

List of Figures vii

List of Tables viii

Acronyms ix

1 Introduction 1
- 1.1 Problem Background 1
- 1.2 Problem Statement 2
- 1.3 Objectives and Scope 3
  - 1.3.1 Aim 3
  - 1.3.2 Objectives 3
  - 1.3.3 Scope 3
- 1.4 Literature Review 4
  - 1.4.1 Digital Citizen Reporting and e-Governance Systems 4
  - 1.4.2 Blockchain for Local Governance Applications 5
  - 1.4.3 Blockchain-Based Citizen Reporting and Complaint Management Systems 5
  - 1.4.4 Permissioned Blockchains and PoA Consensus Mechanisms . 7
  - 1.4.5 Smart Contracts for Public Issue Management and Opinion Polling 8
  - 1.4.6 User Interfaces for Blockchain-Based e-Governance Applications 9
    - 1.4.6.1 Usability Challenges in Decentralized Governance Systems 9
    - 1.4.6.2 Identity Management and Cognitive Load 9

1.4.6.3 Transaction Latency and User Feedback 10
1.4.6.4 Transaction Costs and Abstraction of Blockchain Complexity 10
1.4.6.5 Comparison with Centralized Reporting Platforms 10
1.4.6.6 Accessibility and Inclusivity Considerations 11
1.4.7 AI-Based Content Moderation in Civic Reporting 11
1.4.8 Need for Automated Content Moderation 11
1.4.9 Architectural Approaches for AI Blockchain Integration . . . 11
1.4.10 AI Oracles and Decentralized Oracle Networks 12
1.4.10.1 Techniques for Content Analysis 12
1.4.10.2 Hybrid Human AI Moderation Models 12
1.4.10.3 Ethical Considerations and Content Permanence . 13
1.5 Report Outline 13

2 Background 15
2.1 Blockchain Technology 15
2.2 Permissioned Blockchain and PoA Consensus 16
2.3 Smart Contracts 16
2.4 Decentralized Applications (DApps) 16
2.5 InterPlanetary File System (IPFS) 17
2.6 Zero-Knowledge and Identity Concepts 18
2.7 AI-based Content Moderation 18

3 Methodology and Implementation 19
3.1 Methodology 19
3.1.1 Research Methodology 19
3.1.2 System Requirements Analysis 20
3.1.2.1 Functional Requirements 20
3.1.2.2 Non-functional Requirements 22
3.1.3 Identified System Components 22
3.1.4 Proposed System Architecture 24
3.1.4.1 Overall System Architecture 24
3.1.4.2 Layered Architecture 24
3.1.5 Proposed Workflow 26
3.1.5.1 Citizen Authentication and Identity Workflow . . . 26
3.1.5.2 Overall System Lifecycle Workflow 28
3.1.5.3 Report Submission Workflow 29
3.1.5.4 Community Consensus and Validation Workflow . . 30
3.1.5.5 Authority Resolution and Action Workflow 31
3.1.5.6 Post-Resolution Verification and Appeal Workflow 31
3.1.6 Project Plan and Timeline 32

3.1.6.1 Project Plan 32
3.1.6.2 Project Timeline 40
3.1.7 Expected Outcomes 40
3.2 Implementation 42
3.2.1 Permissioned Blockchain Implementation 42
3.2.2 Smart Contract Architecture and Logic 44
3.2.2.1 The Report Contract 44
3.2.2.2 Ticket Validation and Sybil Resistance 44
3.2.2.3 State Transition Logic 45
3.2.3 Identity and Access Control Implementation 46
3.2.4 Backend Relayer Implementation 47
3.2.5 IPFS Off-chain Storage Integration 49
3.2.6 AI-based Moderation System Implementation 50
3.2.7 Web-based DApp Development 53
3.2.7.1 User Authentication Interface 53
3.2.7.2 Report Submission Interface 53
3.2.7.3 Blockchain Query Interface 54

4 Testing, Experimentation, and Validation 55
4.1 Experimental Setup 55
4.1.1 Hardware Infrastructure 55
4.1.2 Software and Deployment Environment 56
4.1.3 Blockchain Configuration (Server 1) 56
4.1.4 Application and Services Topology 56
4.2 Testing Methodology 57
4.2.1 Subsystem Testing 57
4.2.2 Integration Testing 58
4.2.3 Functional Testing 58
4.2.4 Blockchain Performance Testing 59
4.2.4.1 Transaction Throughput Testing 59
4.2.4.2 Transaction Latency Testing 60
4.2.4.3 Block Confirmation and Finality Testing 60
4.2.5 Smart Contract Validation 61
4.2.5.1 Ticket Validation and Sybil Resistance 61
4.2.5.2 Report Creation and Access Control 62
4.2.5.3 State Transition and Finite State Machine (FSM) Logic 62
4.3 AI Moderation Testing 63
4.3.1 API Availability and Security Tests 64
4.3.2 Text Moderation Classification Tests 64
4.3.3 Oracle Voting Validation 66

4.3.4 Response Structure Validation . . . 66
4.4 IPFS Storage Testing . . . 66
4.5 End-to-End Integration Testing . . . 68
4.6 Comparison with Existing Systems . . . 74
4.6.1 Centralized Reporting Platforms . . . 74
4.6.2 Traditional Content Moderation . . . 74
4.6.3 Public Blockchain Approaches . . . 75
4.6.4 Domain-Specific Blockchain Solutions . . . 75

5 Discussion **76**
5.1 Novel Contributions of the Project . . . 76
5.2 Overall System Validity . . . 76
5.3 Achievement of Proposed Objectives . . . 77
5.4 Summary of Results and Observations . . . 77
5.5 Real-world Applicability . . . 78
5.6 Limitations of the Current Prototype . . . 78

6 Conclusion and Future Work **79**
6.1 Conclusion . . . 79
6.2 Future Work . . . 79
6.3 Final Remarks . . . 80

Bibliography **81**

List of Figures

1.1 System Architecture of the Police Complaint Management System.[1] 6

3.1 High Level Architecture Diagram. 25
3.2 Citizen Authentication and Identity Workflow 27
3.3 Overall lifecycle workflow of a civic issue within the decentralized framework. 35
3.4 Report Submission Workflow 36
3.5 Community Validation Workflow 37
3.6 Authority Resolution and Action Workflow 38
3.7 Post-Resolution Verification and Appeal Workflow 39
3.8 Project Timeline Gantt Chart 41
3.9 Validator node configuration of the private permissioned blockchain. 43
3.10 State transition diagram for the civic reporting system. 45
3.11 Backend relayer workflow architecture. 48
3.12 Oracle network architecture for AI moderation. 51

4.1 Citizen Authentication Interfaces 69
4.2 Report Submission Form 71
4.3 Views of the Submitted Report in the Community Feed 73

List of Tables

3.1 Project plan and current progress 32

4.1 AI moderation API availability and security tests 65
4.2 AI text moderation classification tests 65
4.3 Oracle voting validation 66
4.4 Oracle aggregator response fields 67

Acronyms

AI	-	Artificial Intelligence
API	-	Application Programming Interface
CID	-	Content Identifier
CI/CD	-	Continuous Integration and Continuous Deployment
CNN	-	Convolutional Neural Network
DApp	-	Decentralized Application
FSM	-	Finite State Machine
IPFS	-	InterPlanetary File System
KPI	-	Key Performance Indicator
NGO	-	Non-Governmental Organization
NLP	-	Natural Language Processing
PaaS	-	Platform-as-a-Service
PII	-	Personally Identifiable Information
PoA	-	Proof of Authority
PoS	-	Proof of Stake
PoW	-	Proof of Work
RBAC	-	Role-Based Access Control
RPC	-	Remote Procedure Call
TPS	-	Transactions Per Second
VPS	-	Virtual Private Server
ZKP	-	Zero-Knowledge Proof

Chapter 1

Introduction

Local governance is fundamental to the upkeep of public infrastructure and the overall well-being of the community. To address public problems such as damaged roads, issues in waste management, street light failures, effective communication between citizens and local authorities is critical. Recently, the integration of e-governance systems and crowdsourcing has improve the citizen participation and streamline the reporting and resolution of public issues.
However, most of the current systems rely on centralized architectures. This poses a challenge in dealing with data integrity, transparency, and user privacy. Citizens are reluctant to provide complaints about sensitive topics or criticism due to potential retaliation or the possibility of their personal info being abused. At the same time, the local government is faced with the challenge of verifying the authenticity of anonymous complaints raised by citizens.
Some of the key features of Blockchain technology are immutability, transparency, and decentralization. Those features can be used to address several limitations of conventional reporting systems. It is possible to design reporting platforms that protect user anonymity while ensuring the integrity and accountability of reported information by combining blockchain-based architectures and privacy-preserving mechanisms. This project examines the application of blockchain technology in developing a secure, community-driven reporting framework tailored to local governance contexts.

1.1 Problem Background

In the context of local governance, crowdsourced reporting systems and e-governance platforms are widely adopted. By using those systems, citizens can report issues, provide feedback and participate in decision making. It improves the inclusiveness and the accountability of every citizen. In developing countries like Sri Lanka, these systems have the potential to significantly improve the governance efficiency.

Timely identification and the resolution of public conccerns are the key to governance efficiency.
However, despite their advantages, the conventional e-governance and reporting systems are still highly centralized, making them prone to various threats such as data tampering, illegal access, and single points of failure. In addition, the ability to track behavior down to individual users could serve as a discouraging factor, particularly in reporting delicate matters, in a system that lacks anonymity.
Blockchain-based systems have appeared as a feasible solution for managing shared data in a tamper-proof and transparent manner. The ability of permissioned blockchains to enable controlled participation while maintaining accountability through trusted validators makes them a desirable choice for governance applications. By using smart contracts, reporting workflows and opinion polling can be automated rather than relying on centralized intermediaries.
While blockchain technology offers data integrity and transparency, it still poses challenges related to user privacy, content moderation, and trustworthiness among anonymous participants. These challenges remain open research and implementation problems. Combining blockchain with privacy-preserving identity mechanisms and AI-assisted moderation can help address these limitations and supports the need for a comprehensive framework for secure and reliable community-assisted reporting.

1.2 Problem Statement

The existing frameworks of e-governance and crowdsourced reporting systems for local governance do not provide an integrated mechanism that can guarantee data integrity, user anonymity, and trustworthiness of citizen submitted reports. Centralized architectures are prone to unauthorized access, data tampering, and loss of transparency. At the same time, identity dependent systems discourage citizen engagement due to privacy concerns and fear of retaliation.
Although blockchain has been applied to the context of local government context, many blockchain-based reporting systems still fail to balance anonymity with mechanisms for assessing the reliability and quality of submitted reports. Furthermore, the lack of content moderation and trust evaluation increases the risk of misinformation, and malicious or low quality submissions.
Therefore, there is a clear need for a privacy-preserving, blockchain-based reporting framework that enables anonymous citizen participation while ensuring data immutability, transparent workflows, and reliable evaluation of reported information.
This project aims to address this gap by designing and prototyping a permissioned blockchain-based community-assisted reporting system that incorporates smart contracts, off-chain data storage, and AI-assisted content moderation mechanism

suitable for local governance environments.

1.3 Objectives and Scope

1.3.1 Aim

The aim of this project is to design and prototype a blockchain-based, privacy-preserving, community-assisted reporting system for local governance that empowers citizens to report public issues and provide their opinions in a transparent, accountable, and secure manner.

1.3.2 Objectives

1. Design and implement a permissioned reporting blockchain using a PoA consensus mechanism to ensure the integrity and immutability of citizen-submitted reports.

2. Design and deploy smart contracts to automate public issue reporting workflows, including ticket creation, issue resolution and closure, community voting as feedback on resolved issues, and separate opinion polling mechanisms for public consultation.

3. Develop user-friendly interfaces that enable citizens and authorized government officials to interact with the blockchain for reporting issues and tracking their resolution status.

4. Design and integrate an AI-based content moderation system to detect and filter inappropriate textual and visual content in user-submitted reports.

1.3.3 Scope

1. A prototype of the proposed blockchain will be implemented with a minimum of three full nodes. The PoA consensus mechanism will be used to enable efficient and controlled transaction validation within a permissioned network, while excluding deployment of public, permissionless blockchains or globally distributed ledger infrastructures.

2. User authentication and identity decoupling are managed through a simulated identity service, and the design and implementation of a fully decentralized Self-Sovereign Identity (SSI) blockchain are explicitly excluded from the scope of this project.

3. The integration of AI within the system is limited to content moderation using existing machine learning models, libraries, or external Application Programming Interfaces (Application Programming Interface (API)s), and the development of custom, from-scratch deep learning architectures is not within the scope of this work.

4. Transaction validation authority within the blockchain network is restricted to pre-approved institutional stakeholders, including local government entities and recognized non-governmental organizations, and does not permit anonymous public participation or Proof-of-Work (Proof-of-Work (PoW)) based mining.

5. The project emphasizes functional validation, system evaluation, and performance analysis within a controlled experimental environment, and does not address real-world legal compliance, regulatory adoption, legislative reform, or large-scale integration with existing national government information systems.

6. The user interface development is limited to a responsive, web-based DApp accessible via standard mobile and desktop web browsers, and explicitly excludes the development of native mobile applications (iOS or Android) or cross-platform native frameworks.

1.4 Literature Review

1.4.1 Digital Citizen Reporting and e-Governance Systems

Electronic governance (e-governance) is using the information and communication technologies to provide government services. This includes the support of government to citizens (G2C), government to business (G2B) , government to agencies (G2A) etc. By utilizing the e-governance platforms, everyone can gain access to whole range of public services. Additionally citizens can submit complaints and participate in governance decision making process. [2].
Digital citizen reporting system is one of the many use case of e-governance. "FixMyStreet" is a a such existing platform. It enable citizens to report issues for different issue categories. Sanitation, Traffic and Public Safety are some examples. Previous studies shows that these kind of systems improve governance administrative efficiency, civic participation and also they increase the transparency.[3].
Even though there are many benefits of digital citizen reporting in local governance as described earlier, most of them still use centralized server based architecture. This centralized approach have its own limitations and disadvantages. Susceptibility to data tempering, lack of transparency are the main problems that impact

heavily on public trust. In addition to that, mandatory identity disclosure is discouraging public participation on this kind of reporting systems. Therefore these problems motivate the findings of new decentralized, transparent and immutable architectures that still preserve user privacy with accountability.

1.4.2 Blockchain for Local Governance Applications

Blockchain is a technology that has been popular for its ability to have an immutable, transparent and decentralized ledger for public data. Hence, its a transformative tool for e-governance to address the problems that existing implementations had. The Blockchain allow us to verify the immutability of the records, so its suitable for governance applications where the trust and the accountability is crucial. [4], [5].

Blockchain based land registry system implemented in Cook Country, Illinois is a one real world example of the applicability of blockchain in local governance. They aim to reduce the property fraud by consolidating historical ownership records into a verifiable block in the blockchain. Similar to that Delaware has taken initiative to explore the use of smart contracts that's enable real time asset tracking and automate business fillings. This improves the regulatory compliance and oversights [5].

In addition to administrative efficiency, blockchain has been utilized for citizen centric services, such as secure digital identity management. Estonia has taken a initiative to integrate blockchain across healthcare, taxation and even electronic voting systems. So this can provide a mathematically verifiable audit trail for government data. MyPass project in Austin and the World Food Program's building blocks initiative also use Blockchain technology under the hood to manage encrypted identity and transaction records [4].

Despite the advancements in the blockchain base applications, existing literature highlights the challenges of adoption in governance. Scalability constraints, privacy concerns related to identity data, performance overhead and also the need of the legislative and institutional reforms are some examples. These factors needs to be addressed by carefully selecting blockchain architectures and consensus mechanisms before implementing systems for frequent citizen interactions.

1.4.3 Blockchain-Based Citizen Reporting and Complaint Management Systems

The traditional centralized platforms have data integrity issues, transparency issues and trust issues when it comes to citizen reporting. Past works also indicates that centralized complaint registries often have single point of failures , delayed

responses and public mistrust. Therefore blockchain integrated reporting services can be used to mitigate those issues.
A blockchain based police complaint management system has proposed by Pratheeba [6] contains a modular architecture. It's include security, blockchain, web interface and a hybrid cloud storage. This system ensure immutability of complaint records by cryptographic hash linking. And also it utilize RSA based encryption to restrict the access only to authorized personnel. In Addition to that, Pratheeba used Natural Language Processing (NLP) techniques to automatically categorize complaints in the system based on their priorities.

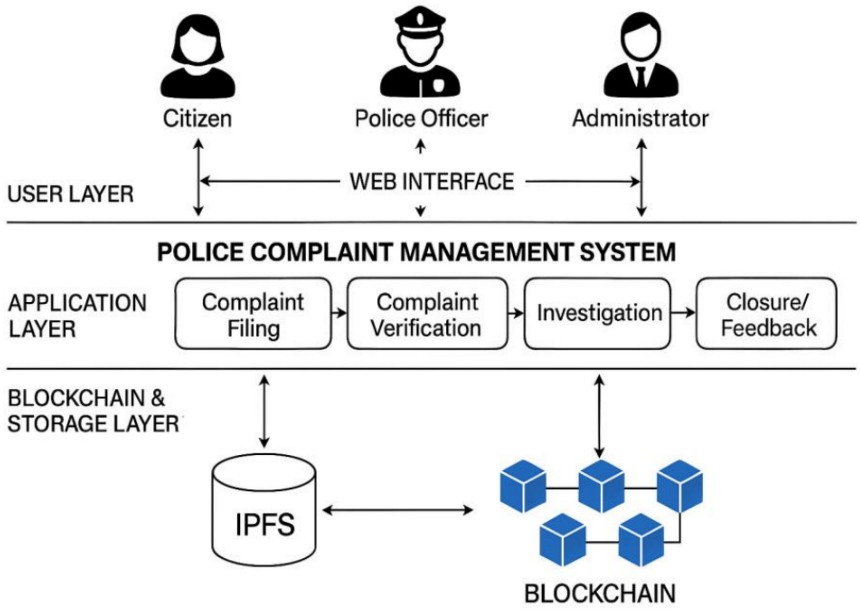

Figure 1.1: System Architecture of the Police Complaint Management System.[1]

A three tier architecture for decentralized complaint management framework has proposed by Manjula [1] is shown in the Figure 1.1. The three layers are web based presentation layer, a business logic layer and finally a blockchain data layer. This system has a role based access control method to involve multiple stakeholders. Citizens, Law enforcement authorities and judicial bodies are some of them. More importantly this system has a immutable storage for multimedia evidences. Intelligent complaint assignment based on workloads, automatic escalation mechanism are some of the key features of the framework. The experimental evaluations shows that a significant reductions in complaint processing time. The successful detection of data tampering also archived by blockchain verification.
In literature, existing blockchain based complaint management systems shows some improvements in transparency and administrative efficiency. But most of them

are domain specific. They primarily target law enforcement or isolated governance functions. In addition to that several platforms still rely on centralized architectures. Only a limited attention is given for implementing a community assisted validation, for privacy preserving reporting mechanisms and efficient handling of multimedia evidences. Therefore these limitations highlight the need for a generalized reporting framework for local governance.

1.4.4 Permissioned Blockchains and PoA Consensus Mechanisms

Based on the participation and access control blockchain network can be classified in to two categories. [7]. They are permissionless architecture (public) and permissioned architecture. Bitcoin and Early Ethereum networks are permissionless blockchains. They allowed unrestricted participation and relyed on consensus mechanism like PoW or Proof of Stake (Proof-of-Stake (PoS))[8]. These architectures and consensus mechanisms emphasized decentralization among participants. But they often suffer from the high latency of the network, limited throughput and unpredictable transaction costs due to congestion. These problems make the traditional permissionless PoW or PoS consensus mechanisms unsuitable for governance and crowdsourced applications[9].
While permissionless blockchains are unsuitable for governance related applications, permissioned blockchains restrict the network participation to authorized entities. This enables more efficient consensus mechanisms and greater control over governance data [7]. This known identities of participants allow permissioned systems to achieve higher performance. And support the regulatory compliances, data privacy and structured governance models as well citemdpi2023unleashing.
Other than PoW and PoS consensus mechanisms, PoA offer significant advantages over permissioned blockchains. PoW requires energy intensive computation. PoS relies on the economic stake and introduce wealth centralization risks. But on the other hand PoA depends on the identity and the reputation of pre approved validators[10], [11]. Therefore this PoA consensus mechanism enables high transaction throughput and low latency. And also it mitigate the issues such as the "nothing at stake" problem by off chain legal and reputational accountability [12], [13].
The government systems should have accountability, performance predictability and data confidentiality. The PoA is well suited consensus mechanism for that [14]. By using PoA we can have the ability to identify and regulate validator entities to align blockchain operations. It's because we have the existing legal frameworks that ensure institutional responsibility. Hence, can avoid malicious behavior [15]. In Addition to that PoA avoids the indeterministic transaction fees and support controlled validation environments. This make them viable for handling sensitive

governmental data under regulatory constraints[11],[16]. As a result, PoA emerges as a practical consensus mechanism for permissioned blockchain-based reporting systems in local governance contexts.

1.4.5 Smart Contracts for Public Issue Management and Opinion Polling

In governance systems, to improve the transparency, efficiency and citizen participation, smart contracts have been explored as the good mechanism. Smart contracts can automate the governance rules and incentives by immutable code. It can support models like civic intelligence governance (CIG). In CIG, citizen engagement is encouraged through on chain participation and token based incentives[17]. Literature shows that approaches like CIG can improve participation rates and knowledge sharing while reducing relances on centralized intermediaries. In the context of multi organizational governance smart contracts also been used within dual blockchain architectures. This method separates transaction data from contract logic. It balance the transparency and privacy in Government to Government and Business to Business processes[18].

Smart contracts also support business processes in governance by embedding policy rules directly into executable logic. Bid submission, evaluation and contract execution are some of the use cases in the public procurement and tendering processes that smart contract hold its value. It also reduce the oppertunities for corruption and increase the accountability[18], [19], [20]. Similar benefits are observed in the blockchain based land registry systems. Smart contracts have enable immutable property records automated ownership transfers and enforcement of land use policies. This has reduced fraud and administrative inefficiencies[21], [22].

In the context of voting and opinion polling, smart contracts offer enhanced security, transparency, immutability and verifiability inherited from blockchain[23], [24]. There are advance mechanisms such as Quadratic Voting and also blockchain enabled participatory budgeting. Those mechanisms further extend citizen involvement in decision making. It have expressive preference representation and transparent allocation of public resources[25], [26]. In addition to these advantages, there exists some challenges related to scalability, privacy and system complexity when deploying voting and polling systems to large and diverse populations[23], [27].

1.4.6 User Interfaces for Blockchain-Based e-Governance Applications

User interfaces play a critical role in determining the accessibility and adoption of blockchain based e-governance systems. In the traditional applications the user literacy is less important but in this conventional blockchain applications the user literacy is highly considered because of the blockchain concepts that are different from the traditional apps and it does make the users hard to understand and adopt. The literature consistently identifies usability and user experience challenges as major barriers to the adoption of decentralized applications for the general public. Particularly where security requirements conflict with established usability principles [28], [29]. So the design of intuitive, inclusive, and responsive web interfaces is essential for enabling effective interaction between citizens, government officials, and the underlying blockchain infrastructure.

1.4.6.1 Usability Challenges in Decentralized Governance Systems

Decentralized applications are fundamentally different from traditional web systems due to its architectural characteristics. Blockchain based systems have asynchronous transactions, immutable records, and cryptographic identity management. They can create friction for users who are used to centralized platforms. Studies highlight a significant usability gap in Web3 applications because of the concepts like transaction finality, gas fees, and private key ownership which are difficult for non technical users to understand and manage [28], [29].

1.4.6.2 Identity Management and Cognitive Load

One of the most critical usability barriers is wallet based identity management. The reason is user identity is tied to cryptographic key pairs instead of institution managed accounts In decentralized systems. Research shows that the responsibility of safeguarding private keys and the unrecoverble key loss red ce the user participation. Specially in general public where inclusivity is essential [29], [30], [31].

The literature further shows that concepts such as self-sovereign identity and identity management remain poorly understood by most users. Studies report unsafe practices and misunderstanding of recovery mechanisms when users are presented with seed phrases or key-based authentication [32], [33]. These findings motivate the need for interface-level abstractions that preserve blockchain security while reducing cognitive load for citizens and officials.

1.4.6.3 Transaction Latency and User Feedback

Blockchain transactions are inherently asynchronous with confirmation times dependent on network conditions and consensus protocols. This latency can lead to confusion among the users because of the system status problems and users can determine those as system failures [29]. The literature recommends to use an optimistic user interface pattern, where user actions are acknowledged immediately while blockchain confirmation occurs in the background.
As advanced design solutions, literature shows progressive status disclosure, distinguishing between off-chain receipt of a report and on-chain verification which allows users to track the lifecycle of an issue without being exposed to unnecessary low level blockchain details [34], [35]. Mechanisms like that are particularly relevant for issue reporting systems that require transparency and trust.

1.4.6.4 Transaction Costs and Abstraction of Blockchain Complexity

A significant usability challenge coming from the blockchain is transaction cost that associated with public blockchains. And also the concepts of gas fees and fluctuating gas costs are very unfamiliar in public services context. Most of the time users expect free access to the reporting platforms[28]. The literature strongly suggest that the use of gas fee abstraction techniques. Meta Transactions is one of the example. In Meta, backend relayer services submit transactions on behalf of the user. Its achieved by preserving cryptographic intent verification[29]. Therefore this approach enable citizen to interact with blockchain without technical or economical complexities.

1.4.6.5 Comparison with Centralized Reporting Platforms

FixMyStreet and SeeClickFIx are existing centralized platforms. Eventhough they are centralized, they provide a valuable benchmarks for crowdsource citizen reporting interfaces. Inituitive map based reporting, simple authentication mechanisms and clear feedback loops are the key characteristics that attributed to platform success[36], [37].
With addition to that SeeClickFix further uses a gamification mechanism to encourage the users to participate in the system. In contrast, Even though the proposed application uses blockchain for reporting platform, it must retain the usability and the expectations of the users. The literature highlights the need of interface design such that gurantees the data integrity and auditability using familiar ui elements rather than cryptographic representations[38], [35], [39].

1.4.6.6 Accessibility and Inclusivity Considerations

Inclusivity is a fundamental requirement for e-governance systems. According to [40], it can be seen that poorly designed blockchain interfaces may digitally divide users. Also it may favor crypto-literate users excluding vulnerable populations. To reduce this issue, literature suggests mobile first design approaches. Furthermore, lightweight client architectures, effiecient bandwidth synchronization can used to support access in resource limited environments. [41].

1.4.7 AI-Based Content Moderation in Civic Reporting

Since blockchain is inheritably immutable its introduce a unique challenge for content moderation in a civic reporting system. In a centralized platforms, administrators can remove or edit inappropriate content. But in a decentralized blockchain based system data cannot be altered once confirmed [42]. While immutability strengthens transparency and accountability, it also raises the risk of permanently storing harmful, illegal, or misleading content. Therefore the literature focuses the need for pre moderation strategies to prevent unsuitable content from being recorded on chain or to limit its visibility after publication [34].

1.4.8 Need for Automated Content Moderation

Civic reporting platforms are susceptible to several forms of misuse that can compromise system reliability and public trust. Automated spam submissions, the submission of malicious or legally sensitive content, and deliberate dissemination of false or misleading reports are some of the common issues [43], [44], [45]. Manual moderation is impractical for above situations. Since the reporting system is large latency, human bias, and operational costs can occur. Furthermore it reintroduces the centralized control structure that blockchain system is aim to minimize.

Therefore the literature supports AI assisted moderation as a scalable and efficient approach for filtering content before its permanently recorded. It maintain the acceptable levels of decentralization [46], [47].

1.4.9 Architectural Approaches for AI Blockchain Integration

Due to the computational and storage limitations of blockchain platforms, AI-based moderation cannot be performed directly on chain. Off chain processing combined with on chain verification identified by existing researches as most practical approach[47], [48].

In this approach, reported content is first analyzed by AI models hosted off chain infrastructure or decentralized compute networks. The outcome of this analysis is a validation decision or a trust score. This outcome then submitted to the blockchain as a cryptographic proof or oracle response. Next the smart contracts verify those results before accepting the report. It will ensure that only the content meeting predefined policies are recorded on chain. BlockCITE is a example framework demonstrate how this separation preserves blockchain integrity while avoiding excessive on chain computation [49].

1.4.10 AI Oracles and Decentralized Oracle Networks

Oracles are trusted intermediaries that connect on chain and off chain. In here the AI oracles provides the computational results to smart contracts. Instead of simply retrieving external data to the blockchain, AI oracle perform content classification and verification tasks. Then it returns the structured results to the blockchain [50], [51].
There can be a risk of oracle bias or failure. To minimize the risk a decentralized oracle networks aggregate outputs from multiple independent AI nodes and determine consensus outcomes. This model reduces reliance on a single AI system. Therefore it improves te robustness by requiring agreement among multiple validators[51], [52].

1.4.10.1 Techniques for Content Analysis

AI-based moderation techniques vary based on the type of submitted content. NLP models are widely used to detect spam, hate speech and misinformation for text based reports. Studies report high accuracy when combining deep learning models with semantic embeddings. It makes them suitable for filtering large volumes of text submissions prior to blockchain recording [43].
For image based reports computer vision models are used to identify inappropriate contents. However recent advances in generative AI increase the risk of fabricated images. This literature proposes a hybrid verification pipeline that combine cryptographic provenance mechanisms with AI-based image analysis. This ensue that the submitted media is both authentic and contextually valid [53].

1.4.10.2 Hybrid Human AI Moderation Models

AI enable the scalable moderation. But it may struggle with contextual interpretation and edge cases. Because of that many studies advocate hybrid moderation models. In those models AI performs initial filtering and prioritization, while human reviewers handle disputed or ambiguous cases [46], [54].

Decentralized community based moderation further complement this approach. The models using community annotation systems, allow users or designated validators to challenge AI decisions during predefined review periods. In Addition to that dispute resolution handled by transparent governance mechanisms [34], [55]. Therefore these strategies reduce the reliance on central authority while maintain the accountability.

1.4.10.3 Ethical Considerations and Content Permanence

Ethical concerns are crucial when implementing systems by integrating AI driven moderation systems. The algorithmic bias and the accountability are the highlights in the literature. Biased training data can lead to unfair rejection of legitimate reports[56]. To minimize this risk explainable AI mechanisms are recommended. This enable users to understand the moderation decision and resubmit corrected reports [57].
Finally, the issue of "toxic immutability" remains a critical challenge for blockch ain-based systems. To resolve this issue its suggested that to use a off chain storage systems like IPFS. If the content is later identified as illegal or harmful, it can removed from the off chain storage. But can preserve the immutable audit trail by recording it on the blockchain[53]. The transparency, legal compliance and ethical responsibility can be balanced by this approach.

1.5 Report Outline

This report is organized into six main chapters. Each chapter focuses on a different part of the project, from the problem background to the current implementation progress and future work.
Chapter 1 introduces the research problem and explains why a privacy-preserving reporting system is needed for local governance. It presents the problem background, problem statement, objectives, scope, and literature review related to citizen reporting systems, blockchain-based governance, AI moderation, and decentralized applications.
Chapter 2 provides the technical background required to understand the proposed system. It explains the main technologies used in the project, including blockchain, permissioned blockchain with Proof-of-Authority consensus, smart contracts, DApps, IPFS, identity privacy concepts, and AI-based content moderation.
Chapter 3 describes the methodology and implementation of the proposed framework. It explains the research approach, system requirements, overall architecture, workflow, project timeline, and expected outcomes. It also describes the implementation progress of the main components, including the private blockchain, smart

contracts, identity service, backend relayer, IPFS integration, AI moderation service, and web-based DApp.
Chapter 4 presents the testing, experimentation, and validation approach. It describes the experimental setup, testing methodology, subsystem testing, integration testing, and preliminary validation of the implemented components. It also discusses the current results and compares the proposed approach with existing reporting systems.
Chapter 5 discusses the main findings of the project. It highlights the important contributions of the proposed methodology and implementation, evaluates the progress against the project objectives, and explains the limitations of the current prototype.
Chapter 6 concludes the report by summarizing the overall progress and significance of the project. It also presents the future work required to complete and improve the system, including stronger multimodal moderation, government-side workflow completion, stronger identity mechanisms, distributed oracle deployment, and further testing.

Chapter 2

Background

This chapter introduces the main background technologies used in the proposed blockchain-based reporting framework. It explains the basic concepts of blockchain technology, permissioned networks, PoA consensus, smart contracts, decentralized applications, InterPlanetary File System (IPFS) , privacy-preserving identity concepts, and AI-based content moderation. Through this chapter we aim to give the reader an enough technical understanding to follow the system design and implementation discussed in the later chapters.

2.1 Blockchain Technology

Blockchain is a distributed, digital, append-only ledger technology. It enables data to be stored in immutable and transparent manner across multiple nodes in a network. Unlike centralized systems, blockchain systems maintain a syncronized copy of the ledger among all authorized participants. This approach reduces the dependency on a single intermediary controlling authority. Each block in the blockchain contains a set of validated transactions, a timestamp, and a cryptographic hash of the previous block. This hash-chaining mechanism ensures that any alteration to previous recorded data becomes immediately detectable.
Immutability is a key characteristic in blockchain technology. It is computationally impractical to alter or delete a transaction after it is confirmed and appened to the blockchain. The immutable property is significantly important in governance-related systems, because integrity and transparency are a must.
Cryptography plays a significant role in blockchain systems. To generate unique digital fingerprints of data, hash functions are used. To verify ownership and authenticity of transactions, digital signatures are used. By using above cryptographic methods, blockchain systems gurantee integirty, authenticity and non-repudiation of data.

2.2 Permissioned Blockchain and PoA Consensus

Bitcoin and Ethereum are permissionless blockchains, where anyone can participate as a node and validate transactions. In contrast, permissioned blockchains restrict participation to a predefined set of entities. These entities are often referred to as validator nodes. They are responsible for validating transactions and maintaining the integrity of the blockchain.
While permissionless blockchains rely on consensus mechanisms like PoW or PoS, the PoA consensus mechanism is specially designed for permissioned blockchains. In here instead of relying on computationally expensive mining or staking process the validator nodes in the permissioned blockchain are validating blocks. Therefore this approach is a reputation and indetity-based mechanism, where the authority of the validators is based on their identity and reputation within the network. This allows for faster transaction processing and lower energy consumption compared to traditional consensus mechanisms. This makes permissioned blockchains more suitable for enterprise applications and private networks.
Local governance related system requires predictable performance, immutability, transparency and accountability. PoA consensus running in a permissioned blockchain is well suited for above requirements because validators identities are known and they controlled by trusted entities. In addition to that, PoA eliminates the need for economic incentives. This make it more efficient and practical for local governance applications.

2.3 Smart Contracts

Smart contracts are first introduced by Ethereum blockchain. They are self-executing contracts with the terms of the agreement directly written into code. In a blockchain based governance systems smart contract can be used to eliminate centralized administrative control. It can automate workflows that would traditionally require human intervention. Therefore this reduce the reliance on intermediaries, hence improve the transparancy and trusthworthiness.
The proposed system uses Solidity based smart contracts deployed on a private permissioned PoA blockchain.

2.4 Decentralized Applications (DApps)

DApps are software applications that operate on a decentralized network, such as a blockchain, rather than being hosted on centralized servers. Unlike traditional web applications that rely on centralized databases and backend infrastructure, DApps utilize smart contracts to handle backend logic and state management. This shift from traditional Web2 to Web3 architecture ensures that application data is immutable, transparent, and resistant to single points of failure or centralized control.

In a typical DApp architecture, the frontend provides the user interface, while the core logic is distributed across the blockchain network. To interact with the blockchain, the frontend utilizes Web3 provider libraries, such as Web3.js or Ethers.js. These libraries enable the application to communicate with blockchain nodes via Remote Procedure Call (RPC). Through this connection, the frontend can query the blockchain to read the current state of smart contracts or construct new transactions. When a user wishes to modify the blockchain state, they must cryptographically sign the transaction using their digital wallet's private key, ensuring authenticity and authorization.

A significant challenge in traditional DApp adoption is the requirement for users to pay transaction fees, commonly known as "gas," using the native cryptocurrency of the network. This creates a barrier to entry for regular users who may not possess or understand cryptocurrency. To address this usability issue, modern DApps employ meta-transactions and relayers. A meta-transaction allows a user to cryptographically sign their intent to execute a transaction off-chain without broadcasting it themselves. This signed message is then sent to an intermediary service known as a relayer. The relayer, which holds a balance of the network's cryptocurrency, wraps the user's signed message in a standard blockchain transaction, submits it to the network on behalf of the user, and pays the associated gas fees. This approach provides a seamless, gasless user experience while maintaining the security and non-repudiation guarantees of cryptographic signatures, making blockchain applications more accessible to the general public.

2.5 IPFS

IPFS is a peer-to-peer network protocol designed for storing and sharing files in a decentralized manner. Unlike the traditional web, where files are identified and accessed through server-based addresses such as URLs, IPFS locates files based on their content. This content-addressed approach improves resilience, reduces dependence on centralized servers, and provides an efficient off-chain storage mechanism for blockchain-based applications. A fundamental concept in IPFS is the Content Identifier (CID), which is a cryptographic hash generated from the content of a file. If the file content changes, the CID also changes, ensuring strong integrity guarantees because users can verify that the retrieved content exactly matches the expected CID. Since identical content generates the same CID, IPFS also supports deduplication and immutable references. In the IPFS network, data can be hosted by any node that chooses to store or pin the content, allowing files to be retrieved from multiple peers using their CID. This decentralized storage model increases availability, improves fault tolerance, and reduces reliance on a single server, although long-term persistence depends on nodes continuing to pin the data. In blockchain-based systems, IPFS is commonly used as an off-chain storage layer because storing large files directly on-chain is costly and inefficient. Instead, applications store only CIDs and essential metadata on the blockchain, while the actual content remains in IPFS. Users can then retrieve the file through its CID and verify its authenticity using cryptographic hashing. Overall, IPFS provides a scalable and cost-effective decentralized storage solution that combines content integrity, distributed hosting, and efficient off-chain data management.

2.6 Zero-Knowledge and Identity Concepts

Zero-knowledge proofs allow a user to prove that they possess sensitive information without actually revaling the information itself.
The identity decoupling is critical requirment because citizens may hesitate to report sensitive problems they have if their identity is linked to the submitted report. Therefore we decouple identity of the citizen by using public-private key pair associated with the citizen. By leveraging this approach the blockchain ledger will record only pseudonymous identifiers, authorization tickets and cryptographic proofs. This ensure no sensitive personal information is ever oublically exposed.

2.7 AI-based Content Moderation

Citizen reporting platforms are vulnerable to spam Submissions, abusive content, misinformation, and malicious reports. Due to immutability of blockchain records, permanently storing harmful content presents significant ethical and operational concerns. Therefore, content moderation techniques are required before reports are recorded on-chain.
AI techniques are widely used for automated content moderation. NLP techniques are used in text moderation tasks such as:toxicity detection, hate speech detection, spam filtering, and sentiment Analysis. For image moderation, computer vision and deep learning approaches are mostly used. Convolutional Neural Network (CNN)s and transformer-based vision models can classify images based on categories like: explicit content, graphic violence, hate symbols, and unrelated media.

Chapter 3

Methodology and Implementation

3.1 Methodology

This section explains the design followed to develop the proposed framework. It describes how the project objectives were translated into system requirements, how the main components were identified, and why a modular prototype-based approach was selected. The methodology also explains the reasoning behind the selected technologies, workflows, and suggested evaluation approaches.

The proposed system is evaluated through controlled subsystem testing and integration testing. Since the project combines blockchain, privacy-preserving identity, off-chain storage, AI moderation, and a web-based DApp, each component is first designed and validated separately. The components are then integrated through defined APIs, cryptographic data formats, and deployment configurations to evaluate the complete report submission workflow.

3.1.1 Research Methodology

This project follows a design-oriented and prototype-based research methodology. The main objective is to design, implement, and evaluate a blockchain-based, community-assisted, privacy-preserving reporting framework for local governance. The system is developed as a functional prototype within a controlled experimental environment rather than as a large-scale production deployment.

The research approach is based on identifying the limitations of existing centralized citizen reporting platforms and designing a decentralized alternative that improves transparency, integrity, anonymity, and trustworthiness. The project combines several technologies, including a private permissioned blockchain, smart contracts, a simulated identity verification mechanism, off-chain storage, AI-assisted content moderation, a backend relayer, and a web-based decentralized application.

The system is developed using an incremental and modular development approach.

First, the functional and non-functional requirements of the system are identified. Based on these requirements, the overall architecture and system components are designed. Each major subsystem is then implemented and tested independently before being integrated into the complete reporting workflow. This reduces implementation complexity and allows each team member to focus on a specific technical component while maintaining compatibility with the overall architecture.
Each subsystem is first validated independently. After subsystem-level testing, the components are connected through defined APIs, blockchain interfaces, cryptographic data formats, and deployment configurations. The main integration focus at the current stage is the end-to-end report submission scenario. In this workflow, a citizen submits a report through the DApp, the report is authenticated and verified by the relayer, the content is moderated by the AI oracle service, accepted content is stored off-chain, and the final metadata is recorded on-chain.
The project also follows an experimental evaluation approach. The prototype is tested using synthetic reports, manually prepared moderation test cases, safe civic images, unsafe image test samples, and controlled integration scenarios. These tests are used to evaluate the correctness of subsystem behavior, component interaction, security checks, content moderation, and report submission flow.
In addition to controlled laboratory testing, the project team has discussed a possible real-world pilot deployment with the project supervisor. Upon receiving the approval from the Faculty Board, the system may be deployed within the faculty environment for reporting issues within the faculty. This potential pilot would provide an opportunity to evaluate the practical usability, privacy expectations, and reporting workflow of the system in a real-world institutional context. This approach follows the phased methodology described in the project proposal, where requirement analysis, system design, blockchain implementation, smart contract development, identity integration, IPFS integration, AI moderation, DApp development, testing, and evaluation are carried out incrementally.

3.1.2 System Requirements Analysis

The system requirements analysis identifies the main functions and quality requirements that must be satisfied by the proposed framework. These requirements were derived from the project objectives, literature review findings, and the expected reporting workflow of a local governance environment. The requirements are divided into functional and non-functional requirements.

3.1.2.1 Functional Requirements

The functional requirements define the main services that the system must provide.

1. **Citizen authentication:** A citizen should be able to authenticate using the simulated GovID identity service.
2. **Ticket issuance:** After successful verification, the identity service should issue a citizen seed and a batch of one-time ticket IDs. These tickets are used to support anonymous but verifiable participation.
3. **Report creation:** A citizen should be able to create a report containing a description, category, location, timestamp, and optional media evidence.
4. **Payload signing:** The DApp should generate a hash of the report payload and sign it using the citizen's local private key before sending it to the backend relayer.
5. **Relayer verification:** The backend relayer should verify the government-signed ticket, citizen signature, payload hash, ticket usage status, and basic file constraints before forwarding the report for moderation.
6. **AI moderation:** The report text and attached media should be analyzed by the AI moderation service before IPFS upload and blockchain submission. The moderation service should detect spam, abusive or threatening text, non-civic content, and unsafe image content.
7. **Off-chain storage:** Accepted report content and media files should be uploaded to IPFS. The returned CIDs and related hashes should be used to verify the stored content.
8. **Blockchain submission:** The backend relayer should submit accepted report metadata, ticket identifiers, content hashes, and IPFS references to the reporting smart contract.
9. **Report retrieval:** The DApp should allow citizens and authorized users to read submitted reports and their current statuses from the blockchain.
10. **Community validation:** Citizens should be able to validate or reject submitted reports using a ticket-based voting mechanism. This supports community-assisted filtering of genuine and false reports.
11. **Government action workflow:** The system should support government-side actions such as issue review, progress updates, resolution updates, and closure.
12. **Post-resolution verification:** Citizens should be able to verify whether a resolved issue has actually been fixed. If the community rejects the resolution, the report should be reopened for further action.

3.1.2.2 Non-functional Requirements

The system also has several non-functional requirements.

1. **Security:** The system should use hashing, digital signatures, ticket validation, role-based access control, secure API communication, nonce validation, and timestamp validation where required.

2. **Privacy:** Citizen identities should be decoupled from on-chain report records. The blockchain should store only pseudonymous identifiers, ticket hashes, metadata, and cryptographic references rather than personally identifiable information.

3. **Integrity:** Report payloads, media hashes, AI oracle decisions, IPFS content, and blockchain records should be tamper-evident using cryptographic hashes and digital signatures.

4. **Availability:** The deployed services should remain accessible during testing, integration, and demonstration.

5. **Performance:** The system should process reports within a reasonable time under prototype workloads. The blockchain, relayer, IPFS, and AI moderation service should be evaluated for practical response times.

6. **Maintainability:** The system should be modular and containerized to simplify deployment, debugging, testing, and future extension.

7. **Auditability:** Important workflow decisions, such as report submission, moderation decisions, vote counts, state transitions, and authority actions, should be logged or anchored through hashes, blockchain events, or signed responses.

8. **Usability:** The DApp should hide unnecessary blockchain complexity from normal users. Citizens should be able to submit reports without manually paying gas fees or directly interacting with low-level blockchain tools.

9. **Scalability:** Large report content and media files should be stored off-chain so that the blockchain remains lightweight and scalable.

3.1.3 Identified System Components

Based on the above functional and non-functional requirements, the proposed system is divided into several modular components. Each component provides a specific function and communicates with other components through defined interfaces.

1. **Web-based DApp for user interaction:** The DApp provides the frontend interface for citizens and government officials. Citizens use it to authenticate, manage issued tickets, create reports, attach media, submit votes, and view report statuses. Government officials use future dashboard interfaces to review validated reports and update resolution progress.

2. **ZKP GovID simulator for identity verification and ticket issuance:** The ZKP GovID simulator acts as the identity verification component in the prototype. It verifies citizen credentials against a simulated registry and issues a citizen seed together with a batch of government-signed one-time tickets. These tickets allow the citizen to participate anonymously in reporting and voting.

3. **Backend relayer for verification and workflow orchestration:** The backend relayer acts as the middleware coordination layer. It receives signed payloads from the DApp, verifies tickets and citizen signatures, enforces file and request constraints, sends content to the AI moderation service, uploads accepted content to IPFS, and submits blockchain transactions on behalf of users. This relayer design provides a gasless experience for citizens.

4. **AI moderation oracle service for content validation:** The AI moderation oracle service analyzes report text and media before permanent storage or blockchain submission. It contains an aggregator and three oracle workers: a Safety Oracle, a Spam and Abuse Oracle, and a Civic Relevance Oracle. The service returns an ACCEPT or REJECT decision together with confidence values, oracle votes, hashes, and an aggregator signature.

5. **IPFS off-chain storage for report content and media:** IPFS is used to store media evidence off-chain. Instead of storing files directly on the blockchain, the system stores only IPFS CIDs and cryptographic hashes on-chain. This reduces blockchain storage overhead while preserving content integrity.

6. **Smart contracts for report lifecycle management:** The Solidity smart contracts manage report submission, ticket nullifier usage, report state transitions, community validation, authority actions, and post-resolution verification. They enforce workflow rules and provide an immutable record of report status changes.

7. **Private permissioned PoA blockchain for immutable record keeping:** The private permissioned blockchain provides the trusted ledger layer of

the system. It uses Proof-of-Authority consensus to support efficient transaction validation by authorized validator nodes. The blockchain stores report metadata, hashes, IPFS references, voting results, and state transitions.

3.1.4 Proposed System Architecture

3.1.4.1 Overall System Architecture

The proposed system uses a hybrid decentralized architecture that combines on-chain and off-chain components. The blockchain is responsible for storing immutable metadata, workflow states, and references to report content. Computationally intensive and large-data operations are handled off-chain through the relayer, AI moderation service, and IPFS. The overall architecture of the system is shown in the Figure 3.1

The DApp acts as the citizen-facing interface. The ZKP GovID simulator handles identity verification and ticket issuance. The backend relayer verifies signed report submissions and coordinates communication between AI moderation, IPFS, and smart contracts. The AI moderation service evaluates report content before storage and blockchain submission. IPFS stores large report data and media evidence, while the smart contracts manage the report lifecycle on the private PoA blockchain.

This architecture supports the main project goal of preserving citizen anonymity while maintaining report integrity and transparency. The use of a relayer also improves usability because citizens do not need to directly manage gas payments or manually submit blockchain transactions.

3.1.4.2 Layered Architecture

The system can be divided into six major layers.

1. User Layer: This layer consists of the citizens and government officials.
2. Presentation layer: This layer consists of the DApp used by citizens and government officials. It provides interfaces for login, report submission, report viewing, and future government-side actions.
3. Identity and authentication layer: This layer consists of the ZKP GovID simulator. It verifies citizens and issues citizen seeds and ticket batches. This separates real identity verification from on-chain activity.
4. Middleware orchestration layer: The backend relayer acts as the coordination layer. It receives signed report payloads, performs validations, calls AI moderation services, uploads accepted content to IPFS, and submits blockchain transactions.

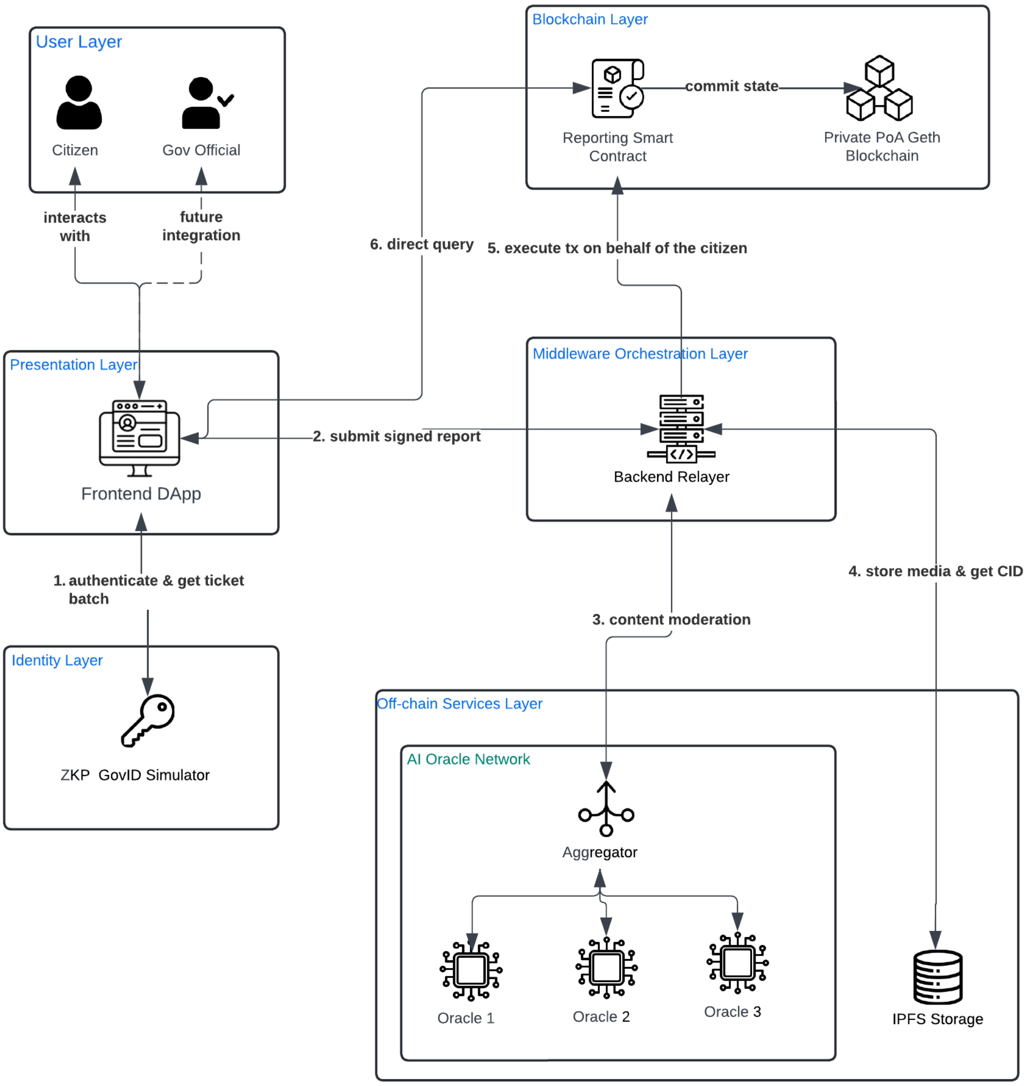

Figure 3.1: High Level Architecture Diagram.

5. Off-chain services layer: This layer includes the AI moderation service and IPFS storage. These components handle tasks that are unsuitable for direct blockchain execution.

6. Blockchain layer: This layer consists of the smart contracts and the private PoA blockchain. It stores immutable metadata, report states, content hashes, IPFS CIDs, and workflow events.

3.1.5 Proposed Workflow

The proposed architecture operationalizes the decentralized reporting framework through a series of distinct, cryptographically secure workflows. These workflows ensure that civic participation remains entirely anonymous to the public and the smart contract, while simultaneously guaranteeing that only verified citizens can submit legitimate data.

3.1.5.1 Citizen Authentication and Identity Workflow

The authentication workflow bridges real-world civic identity with anonymous on-chain participation without exposing Personally Identifiable Information (Personally Identifiable Information (PII)) to the blockchain network. This is achieved through the integration of the ZKP-GovID Simulator, which acts as the centralized trust anchor for identity verification. The workflow is shown in Figure 3.2.
The step-by-step authentication process executes as follows:

1. The citizen accesses the decentralized application frontend and inputs their official national identification number and password.

2. The frontend transmits these credentials over an encrypted channel to the ZKP-GovID Simulator for verification against the mock government registry.

3. Upon successful credential validation, the simulator generates a predefined batch of randomized, single-use ticket identifiers. These identifiers contain no personal data and serve purely as cryptographic tokens.

4. The simulator utilizes the official Government Authority private key to cryptographically sign each randomized ticket identifier.

5. The generated batch of signed tickets, along with a citizen seed value, is returned to the decentralized application and stored securely in the local browser environment. These tickets function as anonymous, verifiable hall-passes for future platform interactions.

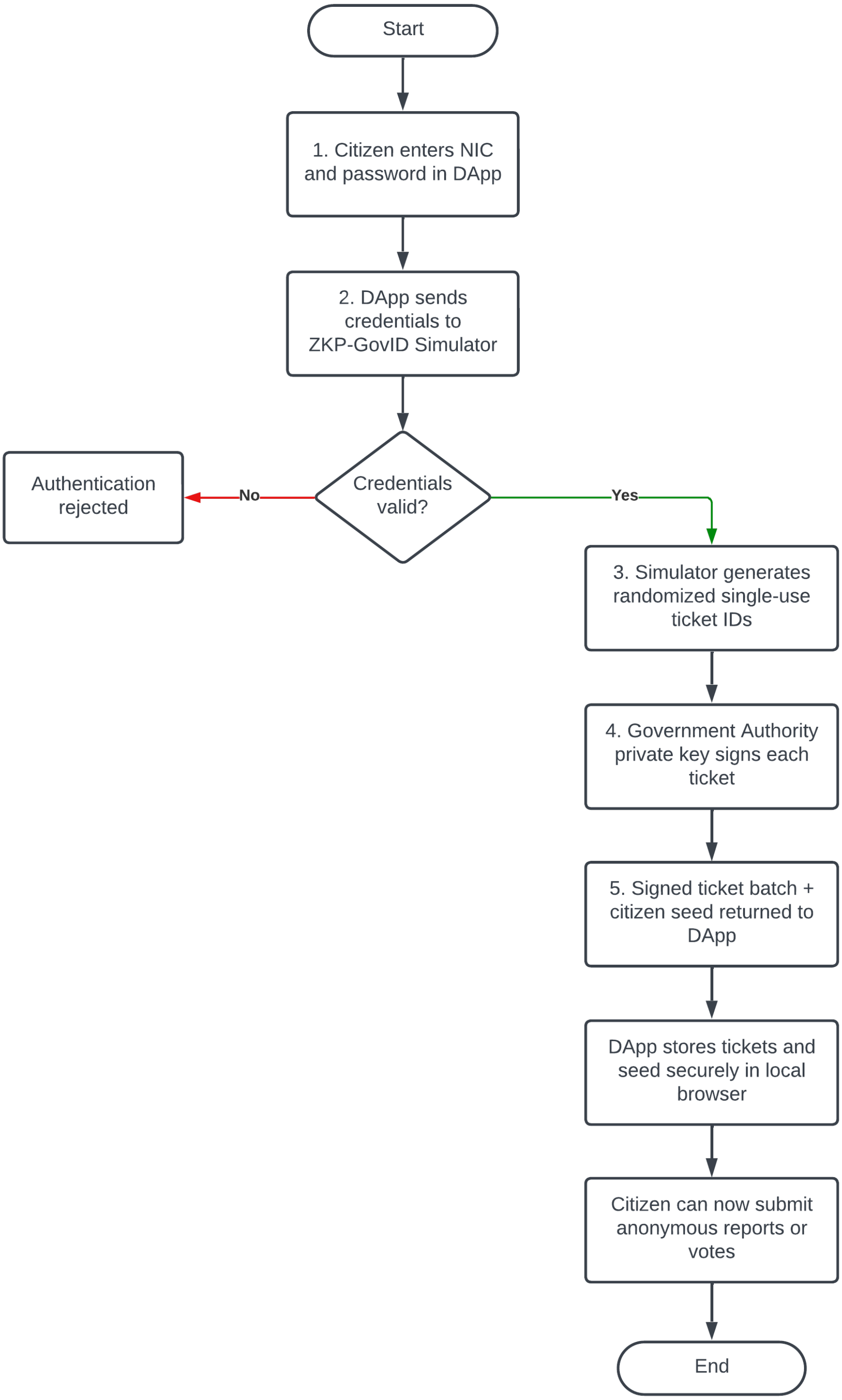

Figure 3.2: Citizen Authentication and Identity Workflow

3.1.5.2 Overall System Lifecycle Workflow

The overall system workflow combines all the separate steps into one complete flow for a civic issue. It tracks a report from the moment a citizen creates it until the government resolves it and the repair is verified. This big-picture view shows how the off-chain middleware, decentralized storage, the AI moderation tool, and the smart contract all work together. The complete lifecycle is shown in Figure 3.3. The progression of a civic report executes sequentially as follows:

1. The lifecycle initiates with the citizen authenticating their real-world identity through the centralized simulator to securely acquire a batch of anonymous cryptographic tickets.

2. Utilizing an acquired ticket, the citizen constructs and submits a civic issue report accompanied by a cryptographic signature.

3. The backend relayer intercepts the payload, routing it through the AI moderation ensemble to filter malicious content, and subsequently uploads the approved media to the decentralized storage network.

4. The relayer formulates the blockchain transaction and anchors the lightweight metadata to the smart contract, placing the report into the initial pending validation state.

5. The local community evaluates the pending report. Reaching the rejection threshold terminates the issue, while reaching the validation threshold transitions the report to an open state, formally notifying the government authority.

6. The assigned authority acknowledges the validated issue and triggers a state transition to indicate that active, physical repairs are currently in progress.

7. Upon concluding the physical intervention, the authority updates the smart contract to mark the issue as solved, which temporarily shifts the report into a pending verification state.

8. The community conducts a final, physical review of the repaired site. An acceptance consensus transitions the report to the terminal closed state, whereas a rejection consensus reopens the issue, mandating the authority to rectify the inadequate repair.

3.1.5.3 Report Submission Workflow

The report submission workflow represents the core operational pipeline of the framework. It delegates heavy computational tasks, such as content moderation and file storage, to off-chain middleware, ensuring the blockchain remains a lightweight and gas-efficient state machine. The workflow is shown in Figure 3.4. When a citizen initiates a civic issue report, the system executes the following deterministic pipeline:

1. The citizen drafts a description of the civic issue and optionally attaches multimedia evidence via the frontend interface.

2. The frontend application selects one unused signed ticket from the local batch. It generates a cryptographic hash combining the issue description and the ticket identifier, and subsequently signs this hash using the citizen's local wallet private key.

3. The application transmits the complete payload, including the raw text, media files, government-signed ticket, and the citizen's signature, to the backend relayer via a secure HTTP request.

4. The backend relayer performs a strict dual-layer cryptographic verification. First, it fetches the Government Authority public key to verify the authenticity of the ticket. Second, it verifies the citizen's signature against the payload hash to detect any tampering during transit.

5. If cryptographic verification succeeds, the relayer routes the submitted text and media to the AI moderation ensemble. The AI moderation ensemble evaluates the report using three oracle workers. The Safety Oracle checks toxic text and unsafe image content. The Spam and Abuse Oracle checks spam and low-quality text patterns. The Civic Relevance Oracle checks whether the submission appears to describe a valid civic issue. The aggregator combines these oracle votes and returns a final ACCEPT or REJECT decision, along with confidence values, report hash, decision hash, and an aggregator signature.

6. Upon receiving an ACCEPT verdict from the AI oracle, the backend relayer uploads the approved report content and media files to IPFS. The IPFS network returns immutable CIDs representing the stored content. If the oracle returns a REJECT verdict, the relayer stops the workflow and the report is not uploaded to IPFS or submitted to the smart contract.

7. The relayer formulates a blockchain transaction containing only the lightweight IPFS CID and the ticket identifier. To ensure a gasless experience for the

citizen, the relayer signs this transaction with its own wallet and pays the necessary network fees.

8. The Reporting smart contract receives the transaction. It immediately checks the provided ticket identifier against its internal nullifier mapping to prevent replay attacks and duplicate submissions.

9. Once the nullifier is marked as consumed, the smart contract successfully anchors the report to the ledger, initializing it in the Pending Validation state to await community consensus.

3.1.5.4 Community Consensus and Validation Workflow

This workflow outlines the mechanism by which the decentralized community filters out spam and validates the legitimacy of submitted civic issues. It ensures that only genuine reports escalate to the government authorities, thereby optimizing resource allocation. The workflow is shown in Figure 3.5.
The validation process executes through the following steps:

1. A newly submitted report enters the Pending Validation state on the smart contract and becomes visible on the community feed within the decentralized application.

2. A registered citizen physically observes the issue in the real world and decides to corroborate the report.

3. The citizen utilizes the application to cast an upvote. To ensure Sybil resistance, the application binds a new, unused Zero-Knowledge ticket from the citizen's local batch to this specific vote and signs the payload.

4. The backend relayer intercepts the voting payload, cryptographically verifies the ticket and signature, and submits the vote to the smart contract.

5. The smart contract verifies that the ticket nullifier has not been used for this specific voting phase of this specific report.

6. The contract increments the vote tally. Once the tally surpasses the predefined validation threshold, the smart contract automatically executes a state transition, moving the report from Pending Validation to the Open state, making it actionable for authorities.

3.1.5.5 Authority Resolution and Action Workflow

This workflow defines how authorized government or non-governmental organization nodes interact with the validated civic issues on the blockchain. It establishes the transparency of the physical repair process. The workflow is shown in Figure 3.6.
The authority interaction proceeds as follows:

1. The government authority monitors the blockchain state via a dedicated dashboard, filtering for reports that have successfully reached the Open state.

2. Upon dispatching a physical repair crew to the location, the authority submits a transaction to the smart contract invoking the state change function.

3. The smart contract validates the transaction signature against its Role-Based Access Control (RBAC) list to ensure the sender possesses the required authority permissions.

4. Upon permission validation, the contract updates the report state from Open to In Progress, providing real-time transparency to the observing community.

5. Once the physical repair is concluded, the authority submits a subsequent transaction to mark the issue as solved.

6. The smart contract processes this transaction and transitions the report into the Pending Verification state, temporarily halting the authority's involvement until the community responds.

3.1.5.6 Post-Resolution Verification and Appeal Workflow

The final workflow enforces the system's core accountability mechanism. It guarantees that government authorities cannot unilaterally close a report without the community physically confirming that the job was completed satisfactorily. The workflow is shown in Figure 3.7.
The verification process executes as follows:

1. Citizens in the vicinity of the purportedly resolved issue inspect the physical location to confirm the repair quality.

2. Citizens utilize the decentralized application to cast a verification vote (accepting or rejecting the repair), utilizing the same secure, ticket-based nullifier mechanism to ensure one-citizen-one-vote integrity.

3. The backend relayer processes and forwards these verification votes to the smart contract.

4. If the positive verification votes reach the required threshold, the smart contract transitions the report to the terminal Closed state, permanently archiving the successful resolution.

5. Conversely, if the negative votes reach the rejection threshold (indicating a poor or non-existent repair), the smart contract transitions the report to the Reopened state.

6. A transition to the Reopened state forces the report back onto the authority's active dashboard, mandating further physical intervention and a repetition of the resolution workflow.

3.1.6 Project Plan and Timeline

3.1.6.1 Project Plan

The project plan is organized according to the main implementation phases of the system. Each phase focuses on a specific part of the proposed framework and contributes to the final integrated prototype. Table 3.1 summarizes the major phases, key activities, expected outputs, and current progress status.

Table 3.1: Project plan and current progress

Phase	**Main Activities**	**Expected Output**	**Current Status**
Background Study and Literature Review	Study existing e-governance systems, blockchain-based reporting platforms, PoA consensus, IPFS, AI moderation, and DApp usability.	Identification of research gap and technical foundation.	Completed
Requirement Analysis	Define functional and non-functional requirements for citizens, government officials, relayer, blockchain, storage, and moderation components.	Requirement specification and use-case understanding.	Completed
System Design	Design overall architecture, component interactions, report submission workflow, identity model, and deployment architecture.	High-level system design and workflow diagrams.	Completed

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Table 3.1 continued

Phase	**Main Activities**	**Expected Output**	**Current Status**
Permissioned Blockchain Setup	Configure private Geth blockchain, PoA consensus, validator nodes, and RPC access.	Working private permissioned blockchain network.	Completed
Smart Contract Development	Implement report creation, report state management, access control, ticket validation, and event emission.	Reporting smart contract for managing issue lifecycle.	In Progress
Identity and Access Control	Implement simulated GovID verification, citizen seed generation, ticket batch generation, and pseudonymous key generation.	Privacy-preserving identity and access control mechanism.	Partially Completed
Backend Relayer Development	Implement payload validation, citizen signature verification, ticket validation, oracle communication, IPFS upload, and smart contract transaction submission.	Middleware service connecting DApp, oracle, IPFS, and blockchain.	In Progress
AI Content Moderation Service	Implement three oracle workers, aggregator, voting logic, relayer request signature verification, text moderation, image safety moderation, decision hashing, and containerized VPS deployment.	Deployed AI moderation service supporting text moderation, image safety moderation, oracle voting, and signed aggregator decisions.	Completed and Integrated with Relayer
IPFS Off-chain Storage Integration	Upload report content and media to IPFS, retrieve CIDs, and verify file integrity using hashes.	Off-chain storage mechanism for report evidence.	In Progress
Web-based DApp Development	Develop citizen login, report submission, report viewing, category selection, and future government-side interfaces.	User-facing web application.	In Progress

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Table 3.1 continued

Phase	**Main Activities**	**Expected Output**	**Current Status**
Subsystem Testing	Test DApp, smart contracts, AI oracle service, GovID simulator, IPFS, and relayer independently.	Verified subsystem behavior.	In Progress
System Integration and Evaluation	Integrate all components and test the full report submission workflow from DApp to blockchain.	Functional end-to-end prototype and evaluation results.	Pending
Final Demonstration	Prepare final system demo, report, and presentation.	Demonstrable prototype and final documentation.	Pending

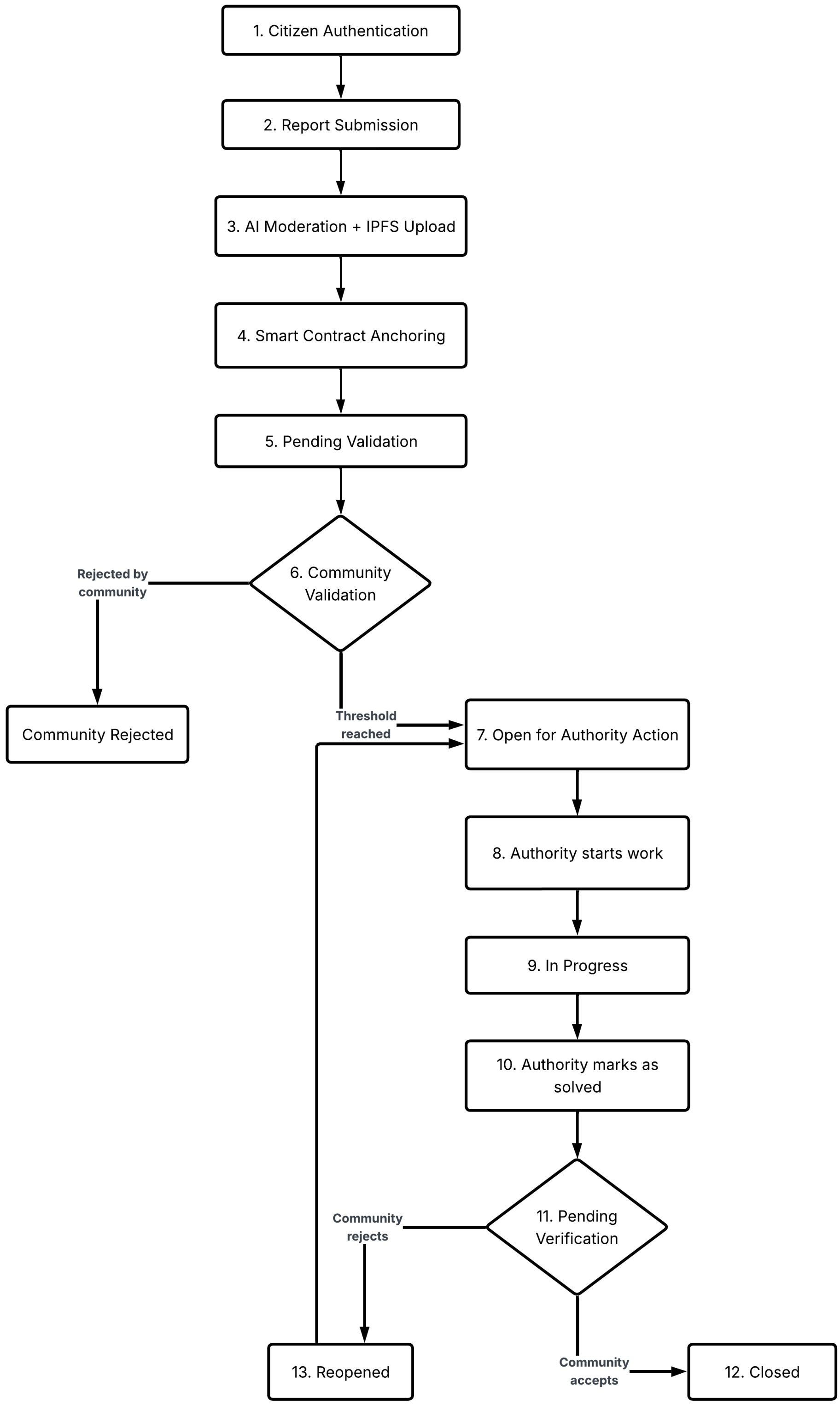

Figure 3.3: Overall lifecycle workflow of a civic issue within the decentralized framework.

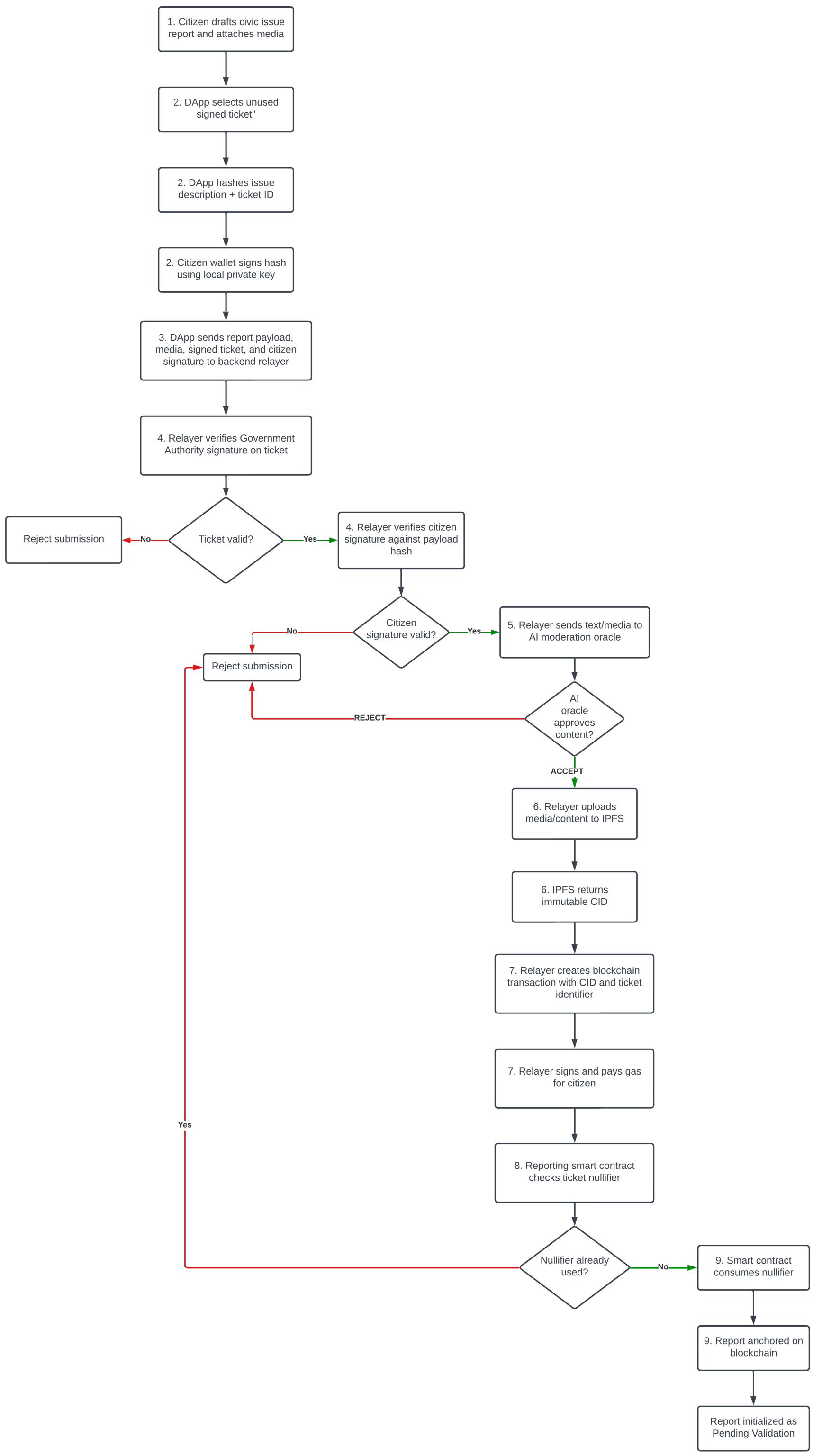

Figure 3.4: Report Submission Workflow

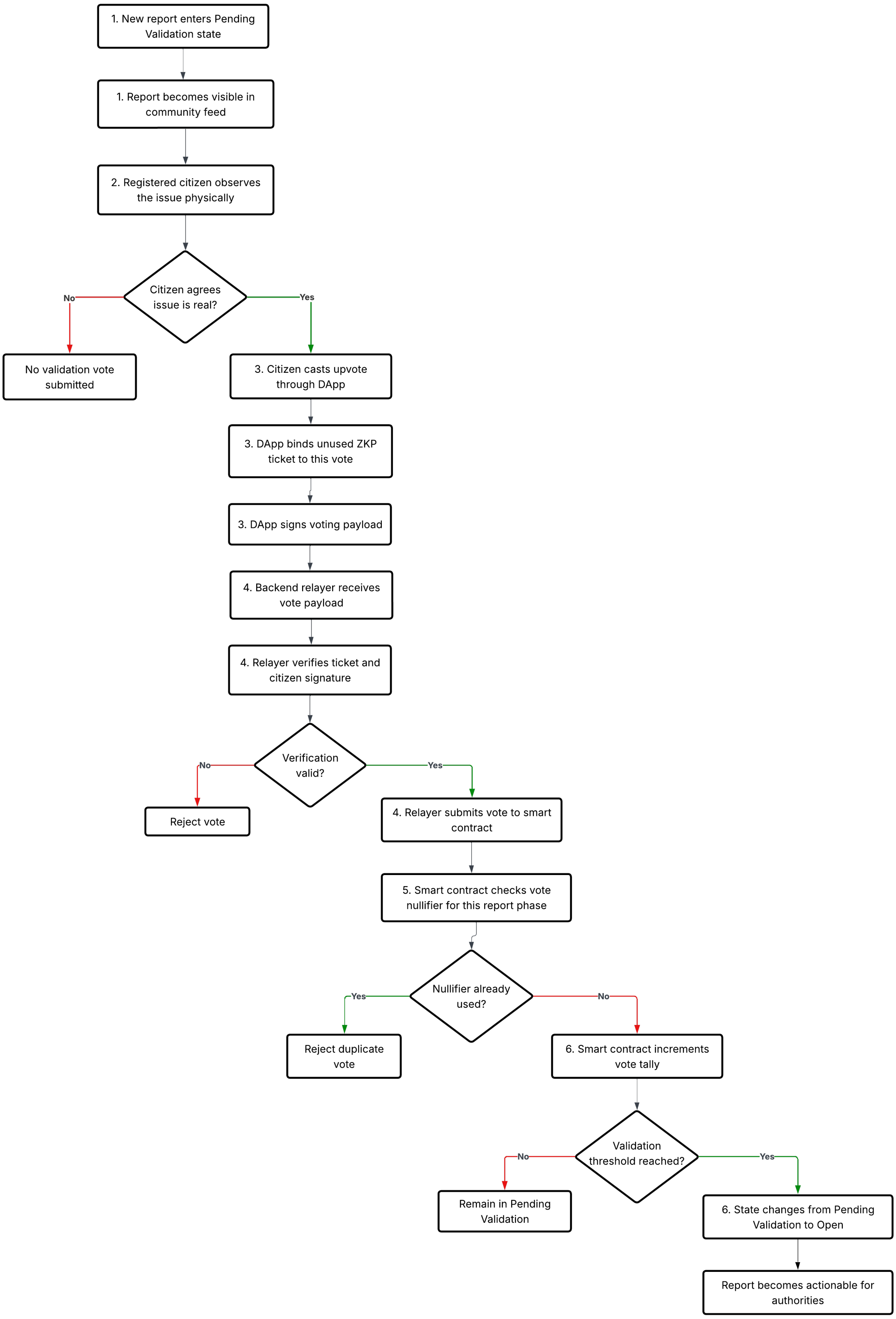

Figure 3.5: Community Validation Workflow

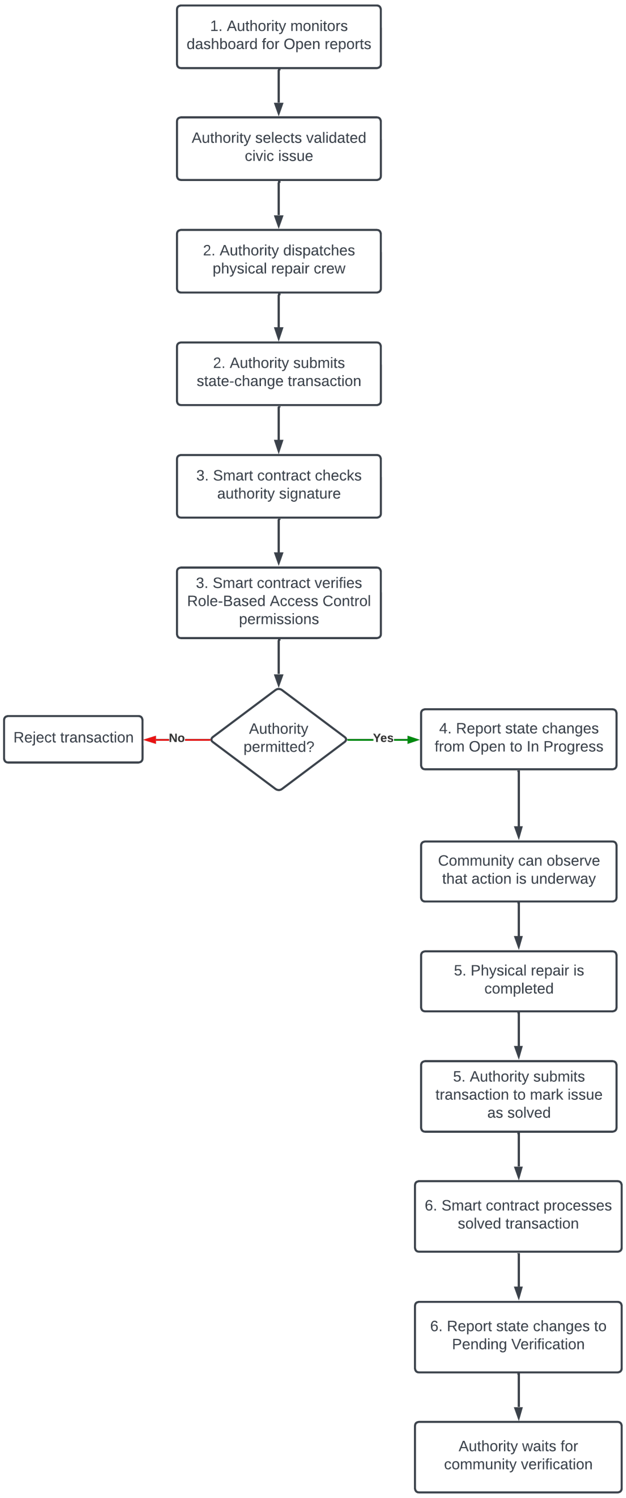

Figure 3.6: Authority Resolution and Action Workflow

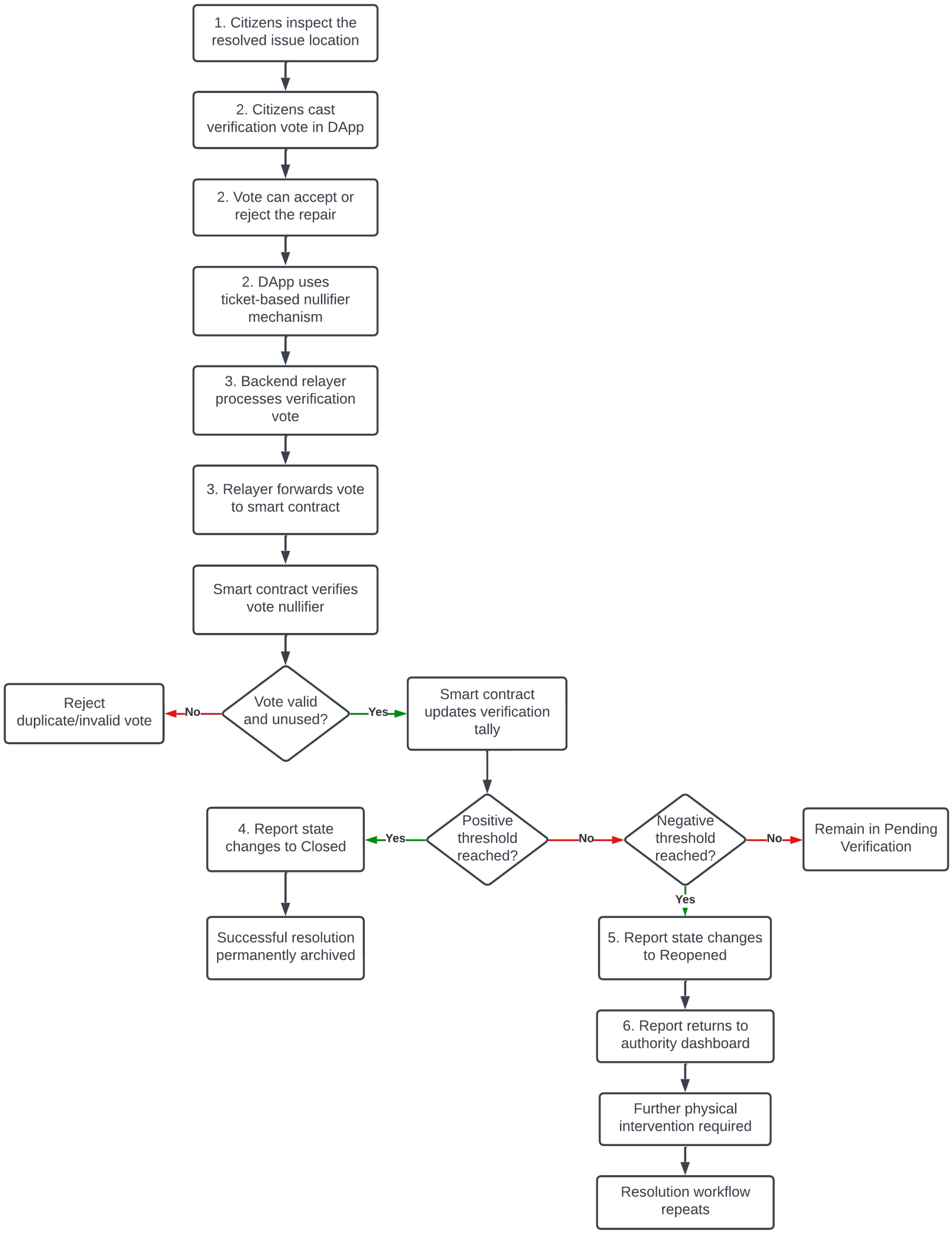

Figure 3.7: Post-Resolution Verification and Appeal Workflow

3.1.6.2 Project Timeline

The project is planned to be completed over two academic semesters using an incremental development approach. The work is divided into several phases, starting from background study and requirement analysis, followed by system design, subsystem implementation, integration, testing, and final evaluation.
The timeline shown in Figure 3.8 was prepared to guide the development process and to allocate sufficient time for each major component of the system. Since this is a progress report, some implementation activities are still ongoing. At the current stage, the main focus is on completing subsystem implementation and moving toward full system integration for the report submission workflow.

3.1.7 Expected Outcomes

The expected outcome of the project is a working prototype of a privacy-preserving, blockchain-based citizen reporting system for local governance.
The system is expected to achieve the following outcomes:

1. Anonymous civic reporting: Citizens can submit reports without directly revealing their real-world identity on-chain.

2. Tamper-evident records: Report metadata, hashes, and workflow states are recorded on a private blockchain.

3. Improved report reliability: AI-assisted moderation helps detect spam, abuse, and irrelevant submissions before blockchain anchoring.

4. Scalable storage: Large report content and media evidence are stored in IPFS, while blockchain stores only metadata and references.

5. Usable blockchain interaction: The relayer pattern removes the need for citizens to directly manage gas fees or blockchain transactions.

6. Modular deployment: The system is designed as a collection of independently deployable components.

7. Governance transparency: The blockchain provides an auditable record of report submission and status changes.

Through these outcomes, the prototype demonstrates the feasibility of combining blockchain, privacy-preserving identity mechanisms, off-chain storage, AI-assisted moderation, and web-based DApp interfaces to support transparent and accountable local governance reporting.

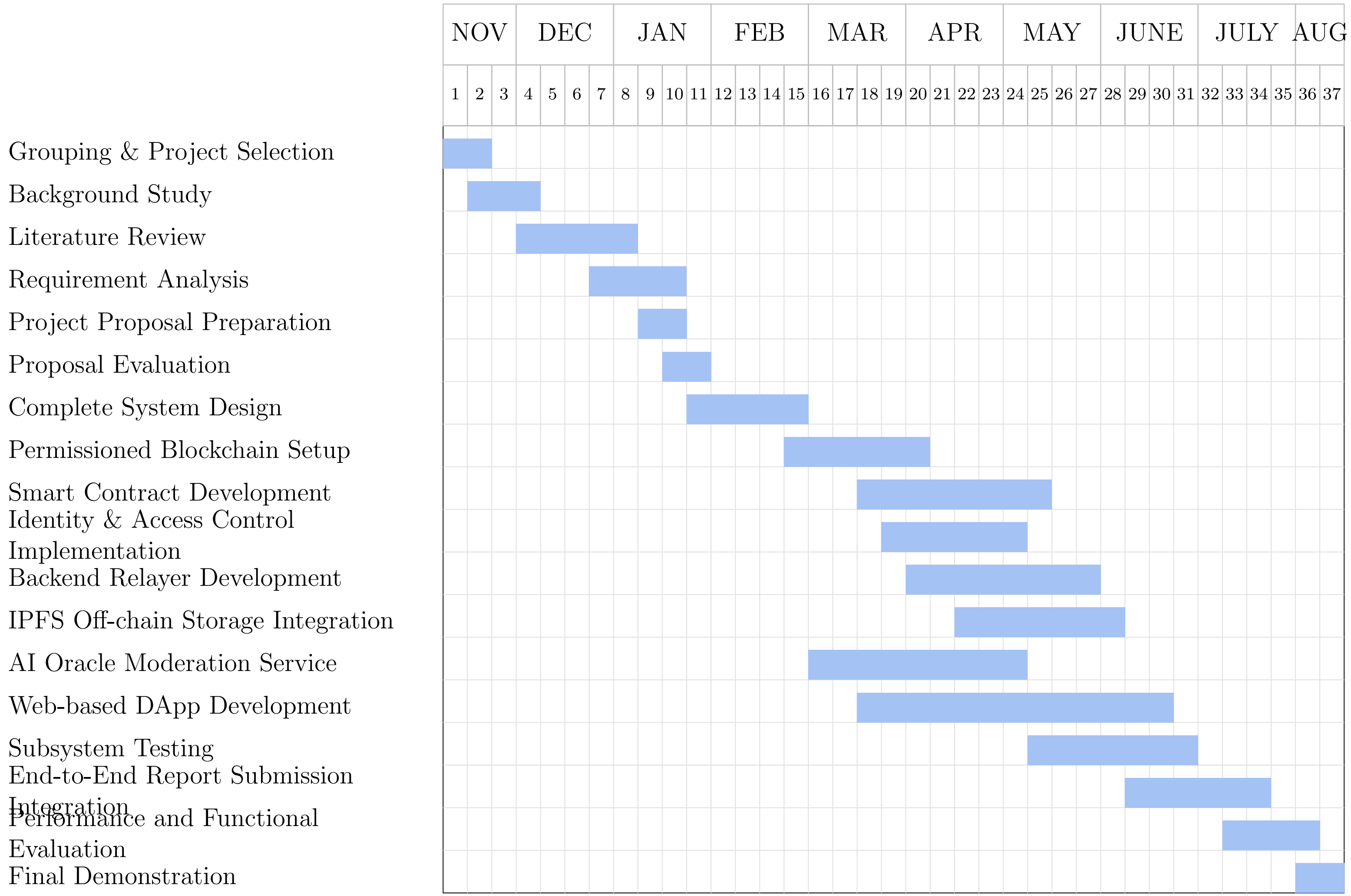

Figure 3.8: Project Timeline Gantt Chart

3.2 Implementation

This section presents the implementation details of the prototype system. The implementation follows the modular architecture introduced in the methodology section, where each subsystem is developed as an independent component and then connected to form the complete reporting pipeline.
The implementation focuses on building a functional prototype that can demonstrate the main reporting workflow. This includes citizen authentication, anonymous ticket-based report submission, backend relayer verification, AI-based text and image moderation, IPFS-based off-chain media storage, and smart contract anchoring on a private permissioned blockchain. The key performance and correctness aspects of these components are later evaluated in the testing and validation chapter.

3.2.1 Permissioned Blockchain Implementation

The core infrastructure of the proposed system is the private permissioned blockchain network. The implementation focused on setting up a secure and efficient environment tailored for local governance applications.

- Geth PoA Network Setup: A private permissioned Ethereum-compatible network was initialized using the Go Ethereum client. The network was configured to operate in proof-of-authority consensus mode, specifically using the Clique algorithm. This enabled faster block confirmation and lower energy consumption compared to traditional proof-of-work systems. A custom genesis file was created to define network parameters, including the initial set of authorized signers and the chain ID.

- Validator Node Configuration: Minimum of three full nodes were required to act as an validator. The 3 nodes are deployed in a personal vps infrastructure. Each node is configured with a unique cryptographic identity and their addresses were added to the list of authorized sealers within the consensus engine to manage transaction validation authority.

- Network Security Measures: To communicate with the blockchain network, secure RPC endpoint was established for node 4. Node 4 is a non-validator node. Additionally, the network is configured to operate in a private subnet, further isolating it from external threats. This configuration allowed DApp and smart contracts to communicate with ledger enabling functions such as report submission, transaction querying and state synchronization.

The node configuration is shown in Figure 3.9.

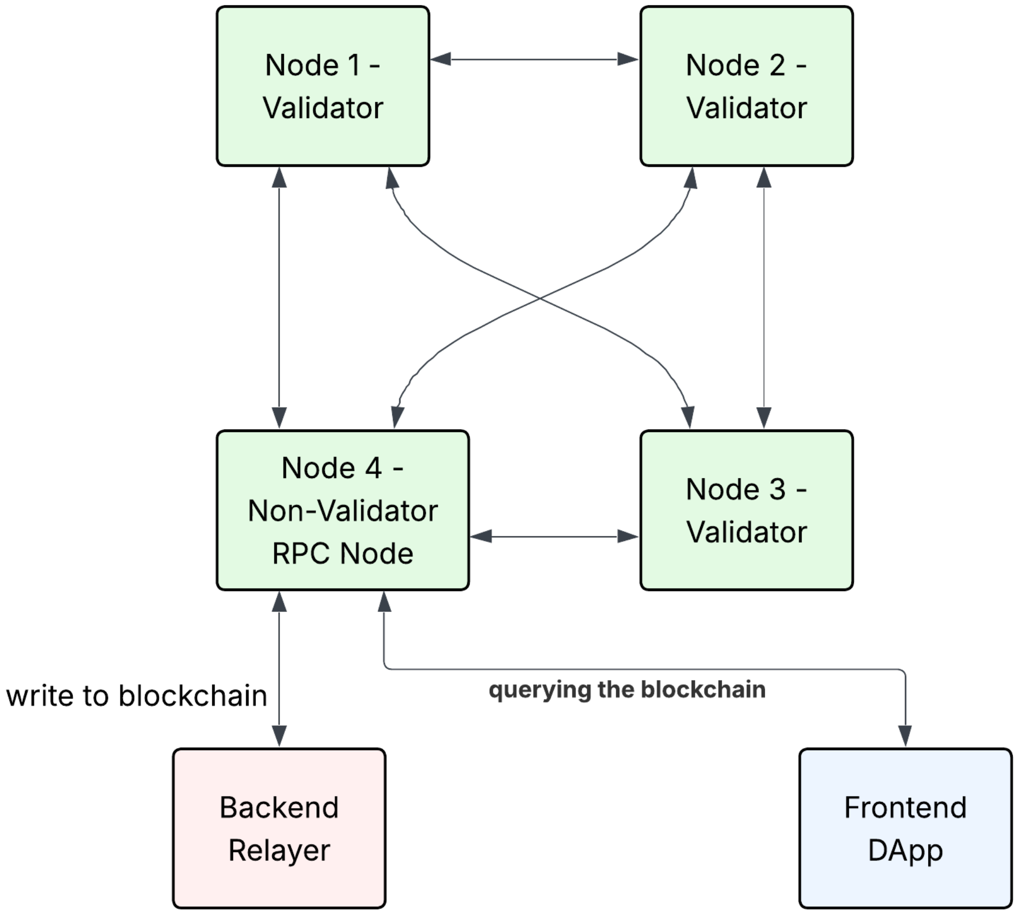

Figure 3.9: Validator node configuration of the private permissioned blockchain.

3.2.2 Smart Contract Architecture and Logic

The core on-chain logic of the decentralized reporting framework is governed by the Reporting.sol smart contract. This contract is designed to be highly gas-efficient, acting primarily as a state machine and a cryptographic ledger rather than a data storage layer. The implementation is divided into three critical functional domains as The Report Contract structure, Ticket Validation, and State Transition Logic.

3.2.2.1 The Report Contract

The Report Contract is engineered using a Separation of Concerns principle. To mitigate the prohibitive gas costs associated with storing large multimedia files on the blockchain, the contract only stores essential metadata.

1. Data Structure: Each civic issue is stored in a Report struct containing a unique integer ID, the IPFS CID pointing to off-chain media, voting counters, the assigned authority address, and its current lifecycle state.

2. RBAC: Utilizing OpenZeppelin's AccessControl, the contract explicitly defines operational boundaries. The RELAYER_ROLE is restricted to the backend server, allowing it to submit reports and relay votes on behalf of citizens. The AUTHORITY_ROLE is granted to recognized government or NGO nodes, granting them permission to update an issue's status to solved or rejected.

3.2.2.2 Ticket Validation and Sybil Resistance

A fundamental challenge in anonymous decentralized systems is preventing Sybil attacks and duplicate submissions. This framework utilizes a dual-layer validation mechanism based on Zero-Knowledge Proof Zero-Knowledge Proof (ZKP) tickets.

1. Off-Chain Cryptographic Validation: Before a transaction reaches the blockchain, the backend relayer intercepts the payload. Using ethers.js, it verifies that the zkpTicketId possesses a valid cryptographic signature from the official Government Authority. It subsequently verifies the citizen's signature against the payload hash to ensure data integrity during transmission.

2. On-Chain Nullifier Tracking: Once validated off-chain, the zkpTicketId is passed to the smart contract as a submissionNullifier. The contract maintains a mapping of public submissionNullifiers. Upon successful report creation, this nullifier is marked as used. Any subsequent attempt to submit a report using the same ticket will systematically fail, cryptographically guaranteeing that one authorized ticket equates to exactly one unique report.

3.2.2.3 State Transition Logic

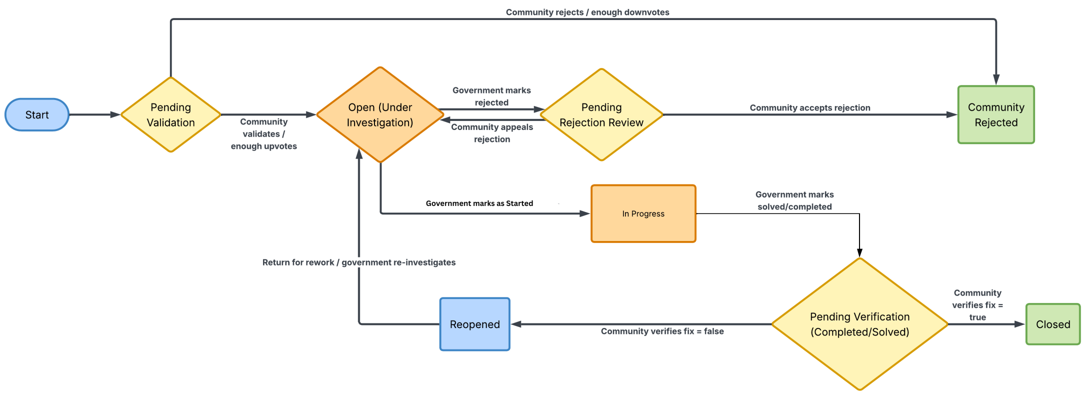

Figure 3.10: State transition diagram for the civic reporting system.

To enforce accountability and prevent unilateral dismissals of civic issues by authorities, the contract implements a strict Finite State Machine (Finite State Machine (FSM)). Every submitted report strictly adheres to an 8-stage lifecycle managed via an on-chain enum:

1. Pending Validation: Initial state awaiting community confirmation of legitimacy.

2. Community Rejected: Terminal state for spam or invalid issues.

3. Open: Validated by citizens and actively awaiting government action.

4. In Progress: The authority has officially acknowledged the issue and is actively executing physical repairs or resolution steps.

5. Pending Rejection Review: The authority dismissed the issue, triggering a community appeal phase.

6. Pending Verification: The authority marked the active work as completed, triggering a mandatory community verification vote.

7. Closed: Terminal state indicating a successfully verified resolution or an accepted rejection.

8. Reopened: The community rejected the authority's fix, forcing the issue back to the authority for rework.

A report must traverse these specific states driven by a combination of authority actions and community consensus thresholds.

1. Community-Driven Transitions: When a report is submitted, it enters the Pending Validation state. The transition to the Open state (making it visible for government action) only occurs automatically via the _changeStatus() internal function once community upvotes reach the predefined VALIDATION_THRESHOLD. Conversely, reaching the REJECTION_THRESHOLD moves it to the terminal Community Rejected state.

2. Authority-Driven Transitions: Authorities can only interact with reports in the Open, Reopened, or In Progress states. An authority calling startWork() shifts an Open or Reopened issue to the In Progress state, providing transparency to citizens that action is underway. From the In Progress state, calling markAsSolved() shifts the report to Pending Verification. Additionally, calling rejectIssue() from the Open or In Progress states shifts it to Pending Rejection Review.

3. Consensus Verification: The FSM mandates that authority actions are never final until verified by the community. If an authority marks an issue as solved, the community must vote to verify the physical fix. Achieving the verification threshold transitions the report to the terminal Closed state. If the community rejects the fix, the state transitions to Reopened, forcing the authority to address the issue again.

3.2.3 Identity and Access Control Implementation

To satisfy the project's requirement for privacy-preserving accountability, an identity decoupling mechanism was successfully developed. This subsystem ensures that citizens can interact with the blockchain anonymously while remaining verified by an off-chain authority. The implementation was structured into three main components

- GovID Simulator: An off-chain simulated identity server was developed and deployed to a personal Virtual Private Server (Virtual Private Server (VPS)). This service acts as the trusted authority for verifying real-world identities. A dedicated API endpoint (/api/govid/verify-citizen) was implemented to securely authenticate users via their National Identity Number (govId) and a password, bridging the gap between traditional web authentication and decentralized access.

- Citizen Keypair generation: To completely decouple a user's real identity from their on-chain footprint, the identity server does not store or transmit blockchain addresses. Instead, upon successful authentication, the simulator generates and returns a secure, 256-bit citizenSeed. This cryptographic seed is then used by the client-side DApp to deterministically generate an anonymous public-private keypair, allowing the citizen to interact securely with smart contracts without exposing personal data.

- Ticket Batch Generation: To facilitate multiple secure, anonymous report submissions over time, the simulator implements a batch authorization protocol. During authentication, the server generates a structured ticketBatch (currently configured to issue 10 tickets per request). Each ticket contains a unique ticketId (a cryptographic hash) and a signature signed by the Identity Authority's private key. Citizens use these single-use tickets to authorize their transactions on-chain. The backend relayer validates the authority's signature on the ticket, granting access without ever knowing which specific citizen is submitting the report.

3.2.4 Backend Relayer Implementation

The backend relayer serves as a critical middleware component that bridges the frontend DApp with the blockchain network and external services. It is responsible for handling report submissions, managing interactions with IPFS for off-chain storage, and orchestrating AI-based content moderation before anchoring data on the blockchain.
The implementation utilizes the NestJS framework to provide a robust and highly modular RESTful API. The core workflow of the relayer executes the following deterministic pipeline for each incoming citizen report.

1. Cryptographic Verification: The relayer receives the payload via a POST /report endpoint. Utilizing the ethers.js library, it performs a strict dual-signature validation:
   (a) It verifies the ZKP Govid Simulator ticket signature against the dynamically fetched Government Authority public key to ensure the user is an authorized citizen.
   (b) It reconstructs the payload hash (binding the description and ticket ID) and verifies the citizen's signature to prevent data tampering in transit.
2. AI Content Moderation: To prevent on-chain spam, toxic immutability of blockchains and ensure platform integrity, the relayer routes the submitted text and media to the AI Oracle moderation service. This service acts as an automated filter, analyzing the content for appropriateness and returning a validation verdict before any storage operations occur.

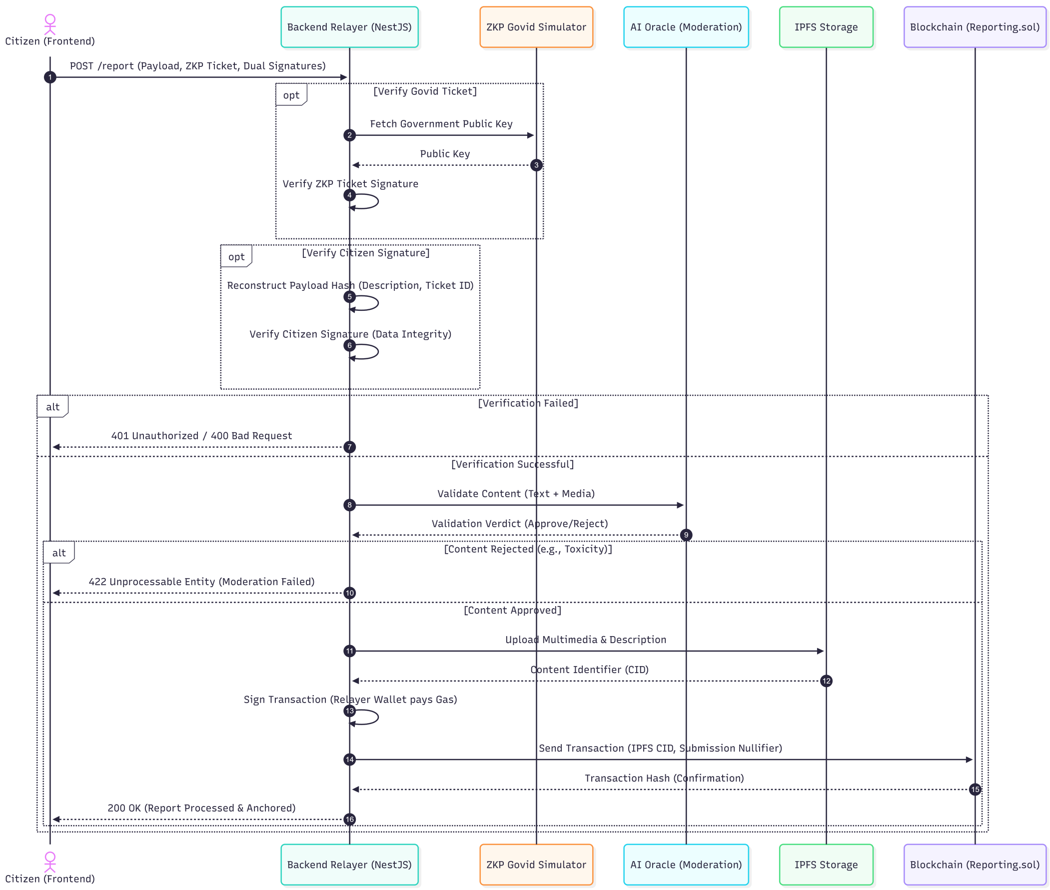

Figure 3.11: Backend relayer workflow architecture.

3. Decentralized Storage (IPFS): To maintain a highly efficient and cost-effective blockchain state, heavy multimedia files and detailed text are offloaded to IPFS. The relayer manages the file buffers via multer, processes the upload, and retrieves an immutable CID.

4. Gasless Blockchain Submission: As the final step, the relayer formats a transaction containing the IPFS CID and the ZKP ticket (which serves as a Sybil-resistant submissionNullifier). The relayer signs and dispatches this transaction to the Reporting.sol smart contract. By paying the network gas fees from its own wallet, the relayer enables a completely frictionless, gasless experience for the end-user.

3.2.5 IPFS Off-chain Storage Integration

The system employs IPFS as the off-chain storage layer dedicated to managing citizen-submitted report images. Rather than recording image files directly on the blockchain, only the Content Identifier (CID) produced by IPFS is passed to the smart contract for permanent on-chain recording. The CID is a cryptographic hash that uniquely represents the exact content of a file, such that any modification to the image produces an entirely different CID. This content-addressing mechanism ensures tamper-evident image storage while significantly reducing on-chain data overhead and improving overall system scalability.

The IPFS integration is implemented as a dedicated microservice structured using a layered architecture comprising route definitions, controller components, utility modules, and middleware layers. The middleware layer enforces file validation by applying image type and size restrictions before any content reaches the storage layer. The controller layer manages the core operations of image upload, CID-based image retrieval, CID verification, and pin management. The utility layer handles direct communication with the IPFS node through the Kubo RPC client.

The image upload workflow is managed entirely by the backend server. When a citizen submits a report through the DApp, the report data and attached image are sent to the backend server. The backend server forwards the image to the AI moderation service for content analysis. If the image passes moderation, the backend server sends only the approved image to the IPFS microservice. The IPFS microservice uploads the image to the IPFS node, which pins the content permanently to prevent garbage collection, and returns the resulting CID to the microservice. The microservice passes this CID back to the backend server, which then combines the image CID with the report data and submits both to the smart contract for permanent on-chain recording.

During retrieval, the image CID is read from the blockchain by the backend server and forwarded to the IPFS microservice, which fetches the corresponding image from the node and returns it for display in the DApp. Since the CID is deterministically derived from the image content, the retrieved image can be independently verified against the on-chain record, confirming that the evidence has not been altered since its original submission.

The service additionally incorporates a CID verification mechanism that confirms a CID is resolvable on the IPFS network before the backend server submits it to the smart contract, preventing broken references from being recorded on-chain. A pin management mechanism is also included to handle image removal where necessary, such as when content is later identified as violating system guidelines. Removing the pin from the local node allows garbage collection to reclaim storage while the on-chain CID record is preserved as an immutable audit trail, directly addressing the toxic immutability challenge identified in the literature. The complete service is containerised using Docker and deployed alongside the IPFS node as a two-container stack, with the IPFS microservice exposed externally on port 4000 and the IPFS node operating internally within the Docker network.

At the current stage of development, the system operates with a single IPFS node within the deployment environment. As a planned future implementation, three IPFS nodes will be deployed across three separate machines and connected as peers within a private controlled network. In this configuration, when an image is uploaded and pinned on the primary node, the two peer nodes will retrieve and independently pin the same content, ensuring redundant availability across all three machines. This multi-node setup will demonstrate the fault tolerance and decentralized storage properties of IPFS at prototype scale, and will operate entirely within the controlled laboratory environment without connecting to the public IPFS network, consistent with the defined project scope.

3.2.6 AI-based Moderation System Implementation

The AI moderation subsystem is implemented as an off-chain oracle service. It is placed between the backend relayer and the IPFS/blockchain submission stage. This ensures that harmful, spam, irrelevant, or unsafe content is filtered before it is permanently stored or referenced by the blockchain.

The moderation service is designed as a containerized oracle network. It contains one aggregator service and three oracle worker services. The backend relayer communicates only with the aggregator. The aggregator then communicates with the oracle workers using internal Docker networking. This keeps the worker services isolated from public access and exposes only one controlled moderation API to the rest of the system. This architecture is demonstrated in Figure 3.12.

The implemented oracle workers are as follows:

- **Safety Oracle:** This oracle checks whether submitted text or images contain unsafe content. For text, it checks toxic, abusive, and threatening patterns.

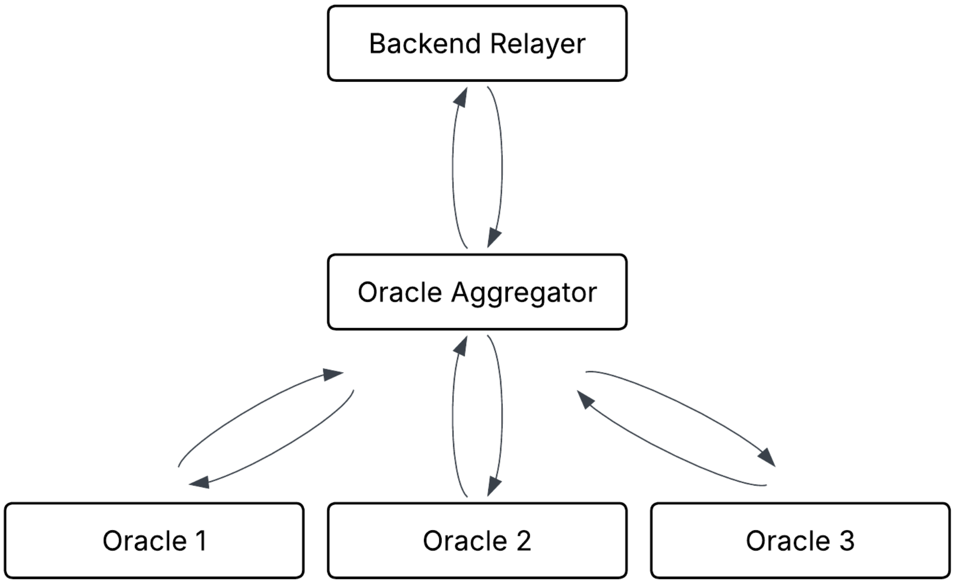

Figure 3.12: Oracle network architecture for AI moderation.

For images, it performs image safety moderation using a pre-trained image classification model to detect unsafe visual content.

- **Spam and Abuse Oracle:** This oracle checks whether the submitted report contains spam-like or low-quality text. It detects promotional phrases, repeated symbols, suspicious URLs, very short descriptions, and common spam patterns.
- **Civic Relevance Oracle:** This oracle checks whether the submitted report is related to a public or local governance issue. It uses category and keyword-based logic to identify reports related to road damage, waste management, drainage, water supply, streetlights, flooding, traffic safety, public property damage, and environmental issues.

Each oracle returns a structured vote containing the oracle identifier, vote, confidence value, explanation code, model or rule version, and additional details. The accepted vote values are ACCEPT and REJECT. The system does not currently use a manual REVIEW state because that would introduce a central human moderation dependency. Instead, unclear or low-quality reports are rejected and the citizen can resubmit a clearer report.

The aggregator combines the oracle outputs using a rule-based decision strategy. If a critical safety violation is detected, the final decision is REJECT. If the civic relevance oracle identifies the content as non-civic, the final decision is also REJECT. Otherwise, the aggregator applies majority voting among the three oracles. If at least two oracle workers accept the content, the final result is ACCEPT. If not, the final result is REJECT.

Security and integrity are also included in the moderation workflow. The backend relayer signs the moderation request using its private wallet key. The aggregator recovers the signer address from the relayer signature and compares it with the trusted relayer address configured in the oracle environment. The request also includes a timestamp and nonce. The timestamp prevents stale requests, while the nonce prevents replay attacks. The aggregator computes a report hash and a decision hash, signs the final decision, and returns this information to the relayer. This makes the moderation result tamper-evident and auditable.

The AI moderation service is deployed on a VPS using Docker containers managed through Dokploy. The deployment consists of the oracle aggregator, safety oracle, spam and abuse oracle, and civic relevance oracle containers. At the current stage, the service has been integrated with the backend relayer and supports both text-based moderation and image safety moderation during the report submission workflow.

3.2.7 Web-based DApp Development

The decentralized application (DApp) serves as the primary gateway for citizens to interact with the local governance system. It is built using a modern web stack comprising Next.js and React for the frontend interface, Tailwind CSS for responsive styling, and Ethers.js for cryptographic operations. The architecture is designed to abstract blockchain complexities away from the end-user while maintaining strict privacy and security guarantees.

3.2.7.1 User Authentication Interface

The authentication interface is designed to verify citizen eligibility without compromising personal privacy.

- **GovID Verification Integration:** The interface securely collects the citizen's GovID credentials and interfaces with an external GovID Simulator API. This abstracts the process of verifying a citizen's real-world identity against a centralized registry.

- **Cryptographic Credential Generation:** Upon successful identity verification, the system retrieves a batch of ZKP tickets and a unique 'citizenSeed'.

- **Local State Management:** Using the 'ethers.js' library, a localized Citizen Wallet is derived from the seed. A custom React Context (`CitizenContext`) is utilized to manage the session state, ensuring that the generated private keys and ZKP tickets remain strictly on the client-side device and are never exposed to external servers. The UI incorporates dynamic feedback mechanisms to indicate the generation of cryptographic proofs to the user.

3.2.7.2 Report Submission Interface

The report submission module allows authenticated citizens to report civic issues anonymously.

- **Form Design and Data Handling:** Developed with a mobile-first approach, the form captures issue descriptions and optional photographic evidence using standard `FormData` structures to support multipart uploads.

- **Anonymous Cryptographic Signing:** To guarantee anonymity and prevent spam, each submission consumes a single ZKP ticket from the user's allocated batch. The payload (consisting of the issue description and the ZKP ticket ID) is hashed using Keccak-256 and subsequently signed using the citizen's locally stored ECDSA private key.

- **Meta-Transaction Relaying:** Instead of submitting transactions directly to the blockchain and incurring gas fees, the frontend dispatches the signed payload, image, and ZKP metadata to a NestJS Backend Relayer. The relayer handles the intermediate steps: uploading the evidence to IPFS, requesting AI moderation, and finally submitting the transaction to the Geth blockchain network on behalf of the user.

3.2.7.3 Blockchain Query Interface

The query interface acts as a community feed where citizens can view, filter, and track the status of reported issues.

- **Mobile-First Feed Architecture:** Implemented as a single-column, scrollable interface inspired by modern social media platforms (e.g., Reddit). This design choice prioritizes readability and user engagement on mobile devices.

- **Dynamic Issue Cards:** Data retrieved from the blockchain is formatted into interactive cards. Each card displays critical metadata including categorical status badges (e.g., PENDING VALIDATION, OPEN, RESOLVED), community upvotes, truncated descriptions, timestamps, and image thumbnails.

- **State Management and Navigation:** The interface employs React state hooks to allow real-time filtering of issues by category (e.g., Infrastructure, Parks & Rec, Safety). Furthermore, a persistent Floating Action Button (FAB) is implemented to provide users with seamless access to the report submission workflow at any time.

Chapter 4

Testing, Experimentation, and Validation

This chapter outlines the proposed methodology for the testing, experimentation, and validation of the decentralized reporting framework. Given the distributed nature of the system—which integrates off-chain cryptographic services, AI moderation, and on-chain smart contracts—a rigorous experimental setup will be established to evaluate performance, security, and functional correctness.

4.1 Experimental Setup

To accurately simulate a real-world deployment of the permissioned blockchain and its ancillary microservices, the experimental environment has been hosted on a cloud-based infrastructure utilizing containerization technologies.

4.1.1 Hardware Infrastructure

The testing environment was deployed across two identical VPS instances to simulate a distributed network architecture and evaluate inter-service communication latency. The hardware specifications for each VPS are as follows:

1. Compute: 4 vCPU Cores
2. Memory: 8 GB RAM
3. Storage: 75 GB SSD
4. Network: High-bandwidth cloud uplink

Server 1 acts as the primary operational node, hosting the core citizen-facing applications, middleware, and the blockchain ledger. Server 2 is strictly dedicated to hosting the computationally intensive AI moderation ensemble and the IPFS storage layer. This isolation prevents resource monopolization and ensures accurate performance profiling of the blockchain consensus mechanism.

4.1.2 Software and Deployment Environment

To ensure reproducibility and consistency across test iterations, the entire software stack was containerized using Docker.

1. Orchestration and CI/CD: Deployment and lifecycle management of the containers are handled by Dokploy, a self-hosted Platform-as-a-Service (PaaS) solution installed on both servers.

2. Automated Deployment: Continuous Integration and Continuous Deployment (CI/CD) pipelines were established via GitHub webhooks. Dokploy was configured to monitor the develop branch of the project repository. Upon receiving a trigger, it automatically pulls the latest source code, builds the respective Dockerfiles for the microservices, and deploys the updated containers.

4.1.3 Blockchain Configuration (Server 1)

The on-chain environment consists of a private, permissioned PoA blockchain.

1. Client Software: The network was initialized using Go-Ethereum (Geth) version 1.13.15.

2. Consensus Engine: The Clique PoA consensus algorithm was utilized to ensure deterministic block times and eliminate the computational overhead of PoW, which closely aligns with the operational model of a government-backed consortium chain.

3. Node Topology: A 3-node cluster was deployed via a unified docker-compose.yml configuration. These nodes serve as the designated Authority (Sealer) nodes representing various governance entities.

4.1.4 Application and Services Topology

The testing setup evaluates the interaction between the following primary subsystems distributed across the two servers.

Deployed on Server 1 (Core Operations):

1. ZKP-GovID Simulator: A Node.js service simulating the issuance of government-backed ZKP tickets.
2. Frontend DApp: A Next.js application providing the citizen interface.
3. Backend Relayer: A NestJS-based middleware component responsible for off-chain cryptographic validation, routing, and gasless blockchain transaction submission.
4. Smart Contracts: Compiled using the Hardhat development environment utilizing Solidity version 0.8.20.

Deployed on Server 2 (Storage and AI Oracle):

1. AI Moderation Ensemble: A multi-node AI oracle architecture designed for robust spam and content filtering. This consists of three distinct AI algorithm nodes running concurrently to analyze incoming media and text. These nodes are governed by a central Aggregator Node that acts as the single entry point and determines the final moderation verdict based on algorithmic consensus.
2. IPFS Node: A dedicated IPFS node that will be deployed in the subsequent testing phase to handle the distributed, off-chain storage of all civic report multimedia (images) and heavy metadata, returning immutable CIDs to the backend relayer.

4.2 Testing Methodology

To ensure the reliability, security, and performance of the decentralized reporting framework, a comprehensive, multi-tiered testing methodology will be employed. This approach systematically validates the architecture from isolated components up to the complete end-to-end user workflow.

4.2.1 Subsystem Testing

Subsystem testing will focus on isolating individual architectural components to verify their internal logic, cryptographic integrity, and algorithmic correctness prior to network integration.

1. Smart Contract Validation: The core reporting smart contract will undergo rigorous unit testing using the Hardhat development framework. Automated tests will be written to validate the logic of the FSM, ensuring that community voting thresholds trigger correct status changes and that role-based access controls prevent unauthorized authority actions.

2. Backend Relayer Verification: The middleware will be tested using a dedicated testing suite. The primary focus will be validating the cryptographic verification algorithms, ensuring that both the ZKP tickets from the simulator and the citizen signatures are correctly authenticated or systematically rejected when malformed.

3. AI Oracle Evaluation: The AI moderation ensemble will be tested in isolation using a curated dataset of synthetic civic reports. This dataset will include a mixture of valid public issues, explicit spam, and manipulated imagery to evaluate the accuracy, false-positive rate, and consensus latency of the multi-node aggregator.

4.2.2 Integration Testing

Once individual subsystems are validated, integration testing will be conducted to evaluate the communication interfaces, network latency, and data flow between the distributed microservices across the two virtual private servers.

1. Relayer and Off-Chain Services: Tests will verify the backend relayer's ability to seamlessly orchestrate the initial validation pipeline. This includes measuring the successful routing of payloads to the AI Oracle ensemble, the handling of timeout exceptions, and the successful uploading of multimedia to the IPFS node to retrieve immutable CIDs.

2. Relayer and Blockchain Interface: Integration scripts will simulate the relayer dispatching transactions to the private PoA blockchain network. This phase will verify the gasless submission mechanics, ensuring the relayer accurately wraps the CIDs and ticket nullifiers into valid transactions that are accepted by the network nodes.

3. Frontend to Middleware Communication: API integration tests will confirm that the decentralized application securely formats and transmits the cryptographic payloads, including the user description and attached media, to the relayer over encrypted channels.

4.2.3 Functional Testing

Functional testing will evaluate the entire integrated system against the defined operational requirements from an end-user perspective, simulating both real-world usage and adversarial conditions.

1. End-to-End User Journeys: Automated scripts will simulate complete, valid report lifecycles. This entails a simulated citizen generating a ticket, submitting a valid issue via the frontend, passing AI moderation, anchoring on the blockchain, and successfully progressing through the community validation and authority resolution states.

2. Sybil Resistance and Replay Attack Mitigation: The framework will be subjected to adversarial functional tests. Scripts will intentionally attempt to submit multiple reports using identical ticket nullifiers to verify that the smart contract successfully rejects duplicate entries and maintains the one-ticket-one-vote invariant.

3. Cryptographic Forgery Attempts: The system will be tested by injecting payloads with manipulated descriptions or mismatched signatures. This will ensure the backend relayer systematically drops tampered requests before they can interact with the off-chain storage or consume blockchain resources.

4.2.4 Blockchain Performance Testing

To quantitatively evaluate the viability and scalability of the proposed decentralized reporting framework, a series of performance benchmarking experiments will be conducted on the 3-node Proof-of-Authority network. These stress tests will be executed using automated benchmarking tools, such as Hyperledger Caliper or custom Node.js scripting, to simulate concurrent user interactions. The evaluation will focus on three primary Key Performance Indicators as transaction throughput, transaction latency, and block confirmation times.

4.2.4.1 Transaction Throughput Testing

The throughput experiment will measure the maximum capacity of the blockchain network to process high volumes of simultaneous civic reports.

1. Experiment Design: The testing suite will generate a progressively increasing load of concurrent report submissions and voting transactions sent to the backend relayer. This mimics a real-world scenario of mass reporting during a localized civic emergency, such as a natural disaster.

2. Significance: High throughput is essential to prevent network congestion, avoid memory pool overflows, and ensure the relayer does not become a bottleneck during peak usage.

3. Metrics and Parameters: The primary metric is Transactions Per Second, calculated by dividing the total number of successfully committed transactions by the total duration of the stress test. Secondary parameters include the transaction success-to-failure ratio and the CPU utilization on Server 1.

4. Thresholds and State-of-the-Art Comparison: Public mainnet blockchains like Ethereum traditionally process between 15 and 30 Transactions Per Second. The benchmark for this permissioned Proof-of-Authority configuration is set at a minimum threshold of 100 Transactions Per Second. Achieving this will demonstrate a highly scalable environment that significantly outperforms traditional public ledgers for civic applications.

4.2.4.2 Transaction Latency Testing

The latency experiments will evaluate the responsiveness of the reporting system from the perspective of the backend relayer and the end-user.

1. Experiment Design: High-precision timestamps will be recorded at the exact moment the relayer dispatches the signed transaction to the blockchain and compared against the timestamp of the finalized block that incorporates that specific transaction.

2. Significance: Low transaction latency ensures a seamless user experience. Extended delays could lead to timeout errors within the middleware interface and cause desynchronization between the off-chain IPFS storage and the on-chain smart contract state.

3. Metrics and Parameters: The core metrics evaluated will be the Average Transaction Latency measured in seconds, and the 95th percentile latency to account for network outliers under heavy load conditions.

4. Thresholds and State-of-the-Art Comparison: While public Proof-of-Stake networks currently target latency periods of 12 seconds or more, the configured Proof-of-Authority consensus mechanism eliminates complex cryptographic puzzles. The expected benchmark for this project is an average latency of under 5 seconds per transaction, representing a substantial optimization over standard public blockchain networks.

4.2.4.3 Block Confirmation and Finality Testing

This phase of testing will analyze the deterministic nature of the Clique consensus engine across the three designated authority nodes.

1. Experiment Design: Network monitoring tools will track the block propagation times across the node topology and measure the block generation intervals during both idle periods and extreme stress-testing phases.

2. Significance: Rapid and deterministic block confirmation is critical for the FSM implemented in the reporting contract. It ensures that community votes and authority state transitions are registered instantly without the risk of chain reorganizations, forks, or orphaned blocks.

3. Metrics and Parameters: The primary parameters are Average Block Time measured in seconds and Time to Finality, which measures the duration required for a transaction to become mathematically irreversible.

4. Thresholds and State-of-the-Art Comparison: Traditional probabilistic finality in legacy networks can take minutes to resolve. For this 3-node Proof-of-Authority setup, the block time parameter is explicitly configured in the genesis block to a fixed interval of 5 seconds. Consequently, the benchmark for absolute finality is expected to be achieved within 1 to 2 blocks, or 5 to 10 seconds. This matches the state-of-the-art performance observed in modern enterprise consortium blockchains.

4.2.5 Smart Contract Validation

To ensure the logical correctness, security, and gas efficiency of the decentralized reporting framework, a comprehensive validation suite will be executed against the core Solidity smart contract. This testing phase will utilize the Hardhat development environment paired with the Chai assertion library to simulate complex on-chain interactions. The validation process is categorized into three critical domains as ticket validation, report creation logic, and state transition integrity.

4.2.5.1 Ticket Validation and Sybil Resistance

This phase will empirically test the cryptographic nullifier tracking system to ensure the network is impervious to Sybil attacks and duplicate reporting.

1. Experiment Design: Automated test scripts will simulate the submission of civic reports using valid, backend-signed ZKP tickets. Immediately following a successful submission, the script will intentionally attempt a replay attack by submitting a secondary report utilizing the exact same ticket nullifier. Additional tests will attempt to inject malformed or mathematically invalid nullifier hashes.

2. Significance: Validating the nullifier mapping is critical to preserving the one-citizen-one-vote invariant. If a malicious actor could bypass this check, they could artificially inflate the severity of an issue or spam the government authority with duplicate tasks.

3. Metrics and Parameters: The primary metric evaluated will be the Transaction Revert Rate under adversarial conditions, alongside the specific Gas Cost required to execute the nullifier validation logic.

4. Thresholds and Benchmarks: The acceptable benchmark for this mechanism is a strict 100 percent transaction revert rate for all duplicate or malformed ticket submissions. Furthermore, the gas overhead for checking the nullifier mapping must remain highly optimized to ensure the relayer is not subjected to excessive network fees.

4.2.5.2 Report Creation and Access Control

These experiments will validate the initialization of civic issues on the blockchain, ensuring that data is stored efficiently and that only authorized entities can write to the ledger.

1. Experiment Design: The test suite will generate transactions attempting to call the report creation function from various network addresses. One address will be properly provisioned with the Relayer Role, while the others will represent unauthorized standard user wallets. The test will verify that only the relayer can anchor the IPFS CID and initial voting metadata to the chain.

2. Significance: This verifies the Separation of Concerns architecture, ensuring the smart contract acts exclusively as a secure state machine rather than a vulnerable, open-access database.

3. Metrics and Parameters: The parameters assessed include RBAC execution accuracy and the Initial State Mapping accuracy. The computational cost will be measured in total execution gas.

4. Thresholds and Benchmarks: Any creation attempt by an address lacking the designated relayer permissions must revert immediately.

4.2.5.3 State Transition and FSM Logic

The most rigorous testing phase will evaluate the strict 8-stage lifecycle of a civic report to guarantee that government authorities and citizens interact within predefined boundaries.

1. Experiment Design: A simulated environment will recreate the complete lifecycle of multiple reports. Tests will incrementally cast community votes to verify that an issue automatically transitions from Pending Validation to Open upon reaching the required mathematical threshold. Subsequently, simulated authority nodes will invoke state-changing functions such as starting work, marking as solved, or rejecting the issue. Finally, the suite will attempt unauthorized state bypasses, such as an authority attempting to manually force an issue into the Closed state without the mandatory community Pending Verification phase.

2. Significance: The FSM is the core accountability mechanism of the project. Validating this logic ensures that authorities cannot unilaterally dismiss public issues and that community consensus dictates the final resolution of every report.

3. Metrics and Parameters: The primary metric is Code Coverage, specifically targeting statement, branch, and function coverage within the smart contract. Secondary metrics include State Transition Accuracy under edge-case voting scenarios.

4. Thresholds and Benchmarks: The testing benchmark mandates 100 percent branch and statement coverage for all state transition functions. The system must yield zero successful unauthorized state bypasses, enforcing absolute adherence to the 8-stage consensus lifecycle.

4.3 AI Moderation Testing

AI moderation testing evaluates whether the moderation subsystem correctly classifies submitted report content as acceptable or unacceptable before IPFS upload and blockchain submission. The implemented AI moderation service now supports both text-based moderation and image safety moderation. Text moderation is used to identify spam, abusive language, threatening content, and irrelevant non-civic reports. Image moderation is used to identify unsafe visual content in attached media files.
The objectives of AI moderation testing are:

1. to verify that the deployed oracle aggregator API is accessible

2. to verify that all three oracle worker containers respond correctly

3. to test acceptance of valid civic reports

4. to test rejection of spam, abusive, threatening, and non-civic text

5. to test rejection of unsafe image content

6. to verify relayer request authentication using API key, timestamp, nonce, and relayer signature

7. to verify oracle voting and final aggregation behavior

8. to verify generation of report hash, request hash, decision hash, and aggregator signature

The AI moderation service is deployed on Server 2 using Dokploy and Docker containers. The deployment includes:

1. Safety Oracle container

2. Spam and Abuse Oracle container

3. Civic Relevance Oracle container

4. Oracle Aggregator container

The backend relayer communicates only with the aggregator API. The aggregator communicates with the three oracle workers using internal container networking. This design prevents direct external access to the oracle workers and keeps the moderation workflow controlled through the aggregator.

4.3.1 API Availability and Security Tests

The first set of tests verifies whether the deployed oracle API is reachable and whether unauthorized requests are rejected. These tests are important because only the backend relayer should be able to call the moderation API successfully. The test cases used for testing moderation API availability are shown in the Table 4.1

4.3.2 Text Moderation Classification Tests

The next set of tests evaluates text moderation behavior. The test cases include valid civic reports, spam reports, threatening content, empty reports, and non-civic content. These tests were executed using an automated signed test client that simulates the backend relayer.
The test cases used for testing moderation classification are shown in Table 4.2

Table 4.1: AI moderation API availability and security tests

Test ID	Test	Expected Result	Status
AI-T01	Call aggregator health endpoint	Service returns running status	Passed
AI-T02	Send valid signed relayer request	Request is processed successfully	Passed
AI-T03	Send request without API key	Request is rejected	Passed
AI-T04	Send request with wrong API key	Request is rejected	Passed
AI-T05	Send request with invalid relayer signature	Request is rejected	Passed
AI-T06	Reuse the same nonce	Replay request is rejected	Passed
AI-T07	Send expired timestamp	Request is rejected	Passed

Table 4.2: AI text moderation classification tests

Test ID	Input Type	Example Input	Expected Decision	Status
AI-T08	Valid civic report	"There is a large pothole near the public school road causing traffic problems."	ACCEPT	Passed
AI-T09	Valid waste management report	"Garbage has not been collected near the public market for three days."	ACCEPT	Passed
AI-T10	Spam content	"BUY NOW FREE MONEY CLICK HERE LIMITED OFFER."	REJECT	Passed
AI-T11	Threatening content	Text containing a public issue combined with a threatening phrase	REJECT	Passed
AI-T12	Non-civic content	"I watched a movie yesterday and the food at the restaurant was very tasty."	REJECT	Passed
AI-T13	Empty report	Empty text field	Validation error / REJECT	Passed

4.3.3 Oracle Voting Validation

The oracle moderation system uses three oracle workers. Each oracle returns either ACCEPT or REJECT. The aggregator combines the votes using a safety-prioritized majority logic. If any oracle identifies a critical safety violation, the final decision is REJECT even if the other oracles accept the report. If there is no critical violation, the aggregator uses majority voting. Table 4.3 summarizes the validation logic.

Table 4.3: Oracle voting validation

Oracle Votes	**Expected Final Decision**
ACCEPT, ACCEPT, ACCEPT	ACCEPT
ACCEPT, ACCEPT, REJECT	ACCEPT
REJECT, REJECT, ACCEPT	REJECT
REJECT, REJECT, REJECT	REJECT
Critical safety violation detected	REJECT

4.3.4 Response Structure Validation

Each successful oracle response should include the fields shown in Table 4.4. These fields support auditability, debugging, and tamper-evidence in the moderation workflow.

The AI content moderation service was successfully deployed as separate Docker containers on a VPS and integrated with the backend relayer. Testing confirms that the aggregator can receive signed moderation requests, verify the relayer signature, communicate with all oracle workers, collect votes, and return an aggregated ACCEPT or REJECT decision. The tests also confirm that unsafe image content can be rejected before IPFS upload and blockchain submission.

4.4 IPFS Storage Testing

The InterPlanetary File System (IPFS) serves as the off-chain storage layer of the proposed system, dedicated exclusively to storing citizen-submitted report images. Since storing image files directly on the blockchain is expensive and impractical, the backend server forwards approved images to the IPFS microservice, which uploads them to the IPFS node and returns the resulting Content Identifier (CID). The CID is a cryptographic hash of the image content, meaning if the image changes in any way, the CID changes as well. This design ensures image integrity without burdening the blockchain with large files. The backend server then submits this

Table 4.4: Oracle aggregator response fields

Field	Purpose
final_decision	Final moderation result returned as ACCEPT or REJECT
final_confidence	Aggregated confidence value of the final moderation decision
risk_level	Indicates the moderation risk level of the submitted report
report_hash	Hash of the submitted report metadata and media hashes
request_hash	Hash of the signed moderation request reconstructed by the aggregator
decision_hash	Hash of the final moderation decision object
aggregator_signature	Digital signature generated by the aggregator over the decision hash
oracle_votes	Individual outputs returned by the Safety, Spam and Abuse, and Civic Relevance oracles
security	Indicates relayer signature verification, timestamp validation, and nonce acceptance
processing_time_ms	Time taken by the oracle aggregator to return the moderation decision

CID along with the report data to the smart contract for permanent on-chain recording.

At the current stage of development, the IPFS microservice has been successfully implemented, containerised using Docker, and deployed to the production server alongside the IPFS node as a two-container stack. The connection between the IPFS microservice and the underlying IPFS node has been established and verified through a health verification test conducted against the deployed service. The test confirmed a healthy status, a live connection to the IPFS node running version 0.41.0, and accurate node information, establishing that the foundation of the off-chain storage layer is stable, deployed, and ready for further integration.

However, since the backend server, the AI content moderation module, and the blockchain layer have not yet been completed, the full testing of the IPFS storage integration is still in progress. The remaining testing activities will be carried out progressively as the dependent components are completed by the respective team members.

Upload performance testing will be conducted once the backend server is ready to forward approved images to the IPFS microservice. Images of different sizes will

be uploaded through the IPFS service layer and the time taken for each upload will be recorded and analysed to assess the responsiveness of the system under realistic reporting conditions.
Retrieval integrity testing will be performed to confirm that images downloaded from IPFS remain identical to the originals. After uploading an image and obtaining its CID, the same image will be retrieved using that CID and compared against the original content byte by byte to verify that nothing was lost or altered during storage and retrieval.
CID validation testing will be carried out to ensure that the generated CID accurately represents the corresponding image content. For each uploaded image, the CID produced by IPFS will be recorded, and after retrieval, the CID will be recalculated from the downloaded content and compared against the original value to confirm that the content-addressing mechanism functions correctly.
AI-assisted content moderation integration testing will be performed once the backend server connects its moderation pipeline to the IPFS microservice. These tests will verify that only images approved by the AI moderation service are forwarded to and stored on the IPFS node, ensuring that inappropriate content is rejected before any CID is generated or passed to the smart contract.
End-to-end integration testing between the IPFS microservice and the deployed smart contracts will be conducted once the blockchain layer and backend server are completed. These tests will verify that image CIDs recorded on-chain correctly resolve back to their corresponding images when fetched through the IPFS retrieval layer, confirming that the complete flow from image submission through backend server, IPFS storage, and on-chain recording operates as intended.
Finally, performance evaluation under simulated load will be carried out to measure upload throughput and retrieval latency when multiple concurrent image submissions are processed simultaneously, as outlined in the evaluation metrics defined in the methodology chapter. Additionally, the planned expansion to a three-node IPFS peer network across three separate machines will be validated to confirm redundant image availability and fault tolerance within the controlled laboratory environment.

4.5 End-to-End Integration Testing

To ensure the seamless operation of the entire platform, end-to-end integration testing will be conducted by simulating the complete lifecycle of a civic issue report. This testing phase will verify the interoperability between the frontend DApp, the backend relayer, the IPFS storage network, the AI moderation oracle, and the underlying permissioned blockchain.
The end-to-end testing workflow will be executed as follows:

1. Citizen Authentication and Key Generation: The test will commence with a simulated citizen authenticating via the external GovID simulator. Upon successful verification, the system is expected to generate a batch of ZKP tickets and derive the local wallet, thereby validating the correct functioning of the identity decoupling mechanism.

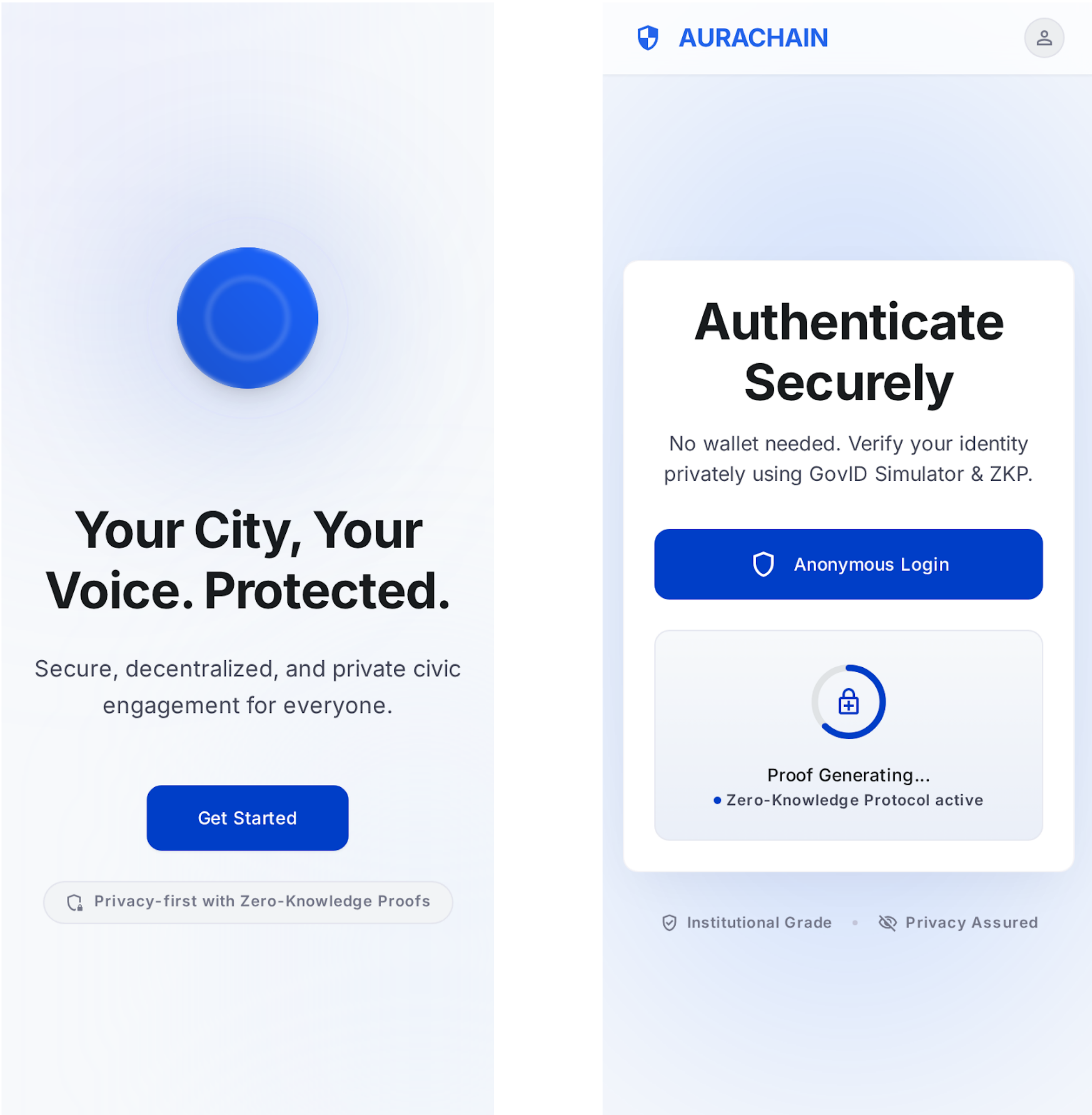

Figure 4.1: Citizen Authentication Interfaces

2. Report Submission and Cryptographic Signing: The authenticated user will submit a test report containing a text description (e.g., "Pothole on Main Street") and a sample image. The DApp will be expected to consume a

ZKP ticket, hash the payload, and sign it using the local private key before transmitting the multipart form data to the backend relayer.

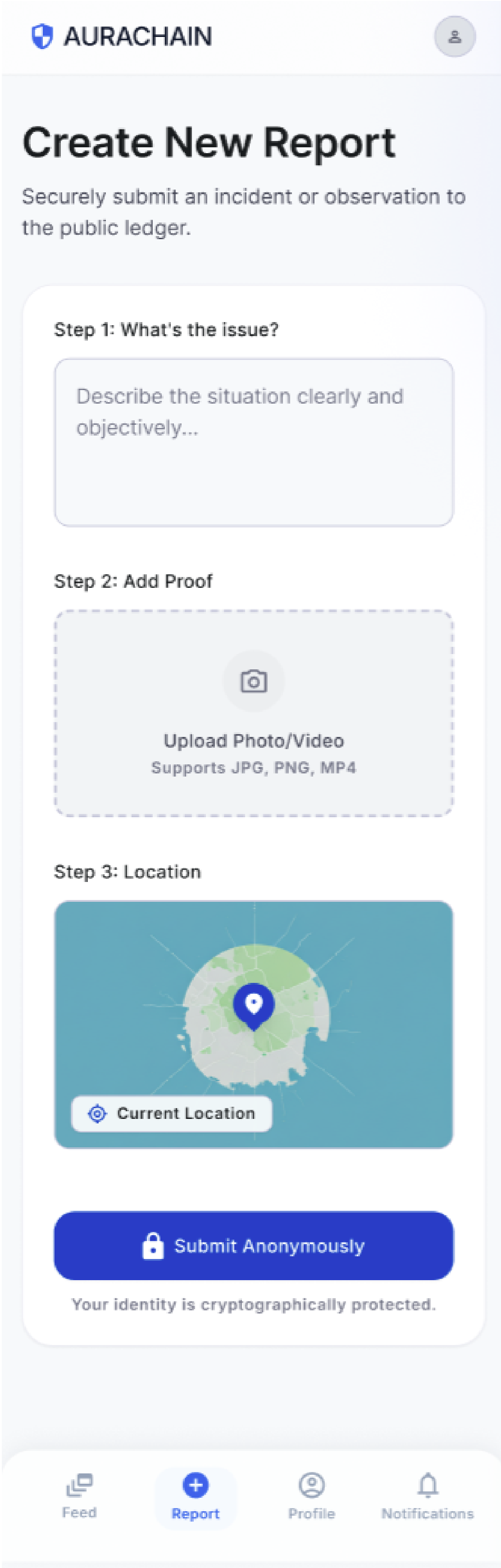

Figure 4.2: Report Submission Form

3. Off-chain Storage and AI Moderation Routing: Upon receiving the payload, the backend relayer will process the submitted data. The visual evidence will be routed to the IPFS integration module, and a CID will be generated. Simultaneously, the text and image data will be evaluated by the AI moderation service to verify that the report is classified as legitimate, non-abusive content before proceeding to blockchain submission.

4. Blockchain Recording via Meta-Transaction: Following AI approval, the backend relayer will format the data, including the IPFS CID, issue description, and the cryptographic proofs. Then submit a meta-transaction to the smart contract on the Geth PoA network. The transaction is expected to be verified by the validator nodes, securely anchoring the report to the immutable ledger without requiring the citizen to pay any gas fees.

5. Frontend Verification: Finally, the DApp's query interface will be refreshed to retrieve the latest blockchain state. The expected outcome is that the newly submitted report will appear in the community feed with its corresponding status badge, upvote count, and image thumbnail, confirming that the entire decentralized pipeline has executed correctly.

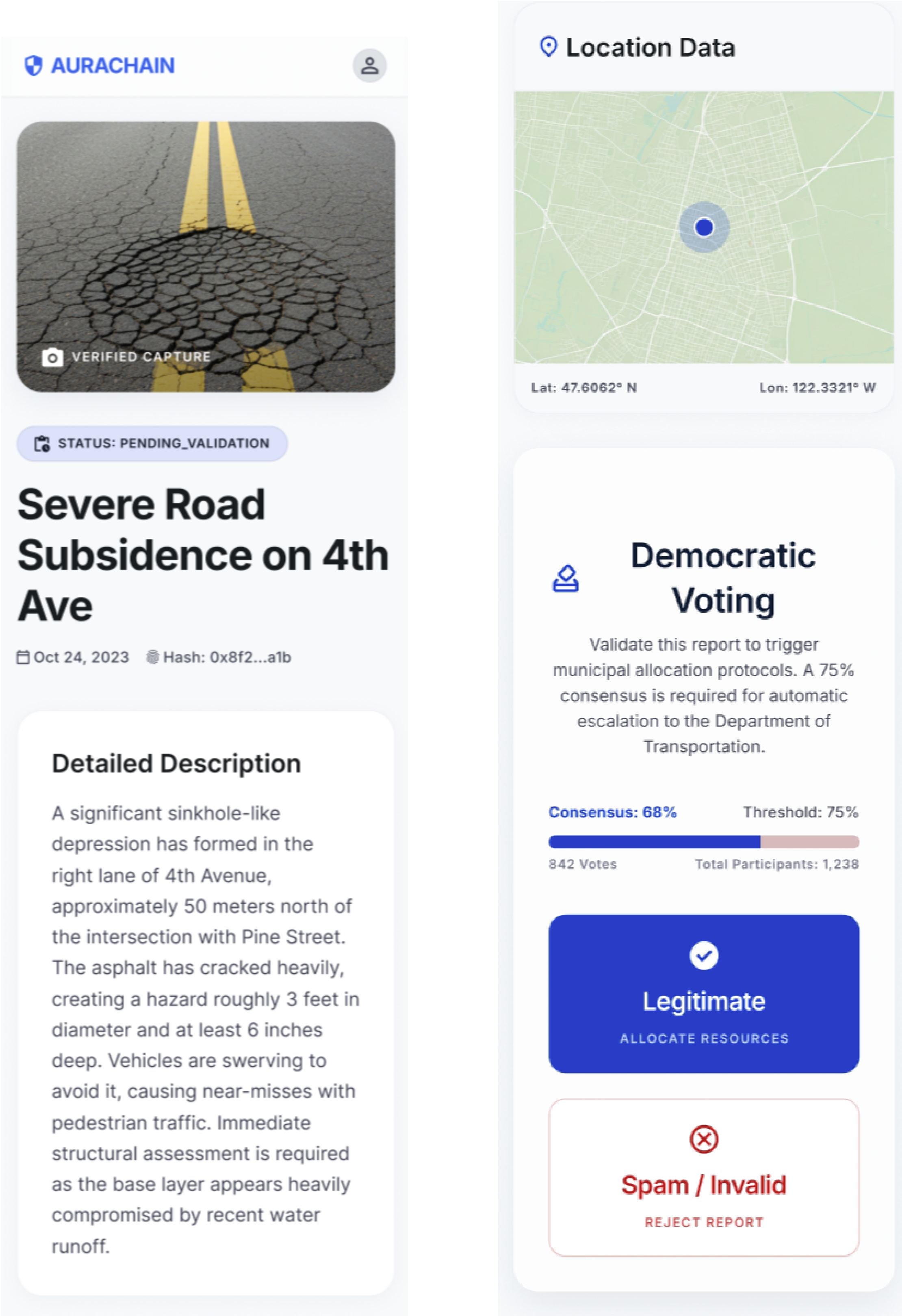

Figure 4.3: Views of the Submitted Report in the Community Feed

The successful execution of this simulated workflow will validate the system architecture and confirm that the individual components interact correctly to provide a secure, anonymous, and user-friendly civic reporting platform.

4.6 Comparison with Existing Systems

To demonstrate the advantages and novel contributions of the proposed community-assisted reporting framework, it is essential to compare its architecture and features against existing solutions in the domain of e-governance and citizen reporting.

4.6.1 Centralized Reporting Platforms

Traditional crowdsourced reporting systems, such as FixMyStreet and SeeClickFix, have successfully established user-friendly interfaces with intuitive reporting workflows and clear feedback loops. However, as identified in the literature, these platforms rely entirely on centralized server architectures. This centralization creates single points of failure and makes the systems susceptible to data tampering and unauthorized modifications. Furthermore, centralized platforms often require mandatory identity disclosure, which can discourage citizen participation, especially when reporting sensitive issues or criticizing local authorities. In contrast, the proposed system leverages a permissioned blockchain to ensure data immutability and absolute transparency. By employing ZKP and identity decoupling, the proposed framework guarantees citizen anonymity while maintaining mathematical accountability, a feature completely absent in conventional centralized systems.

4.6.2 Traditional Content Moderation

Traditional content moderation in centralized reporting platforms is usually performed by administrators or centralized moderation teams. This approach can be effective, but it introduces human delay, moderation bias, operational cost, and single points of control. It also conflicts with the privacy-preserving and decentralized goals of the proposed framework.

The proposed system uses an off-chain AI oracle moderation service instead of a manual centralized moderation queue. The moderation process is automated and takes place before IPFS upload and blockchain submission. The use of three oracle workers reduces dependence on a single moderation model. The aggregator also returns structured outputs, including oracle votes, confidence scores, hashes, and a digital signature. This provides better auditability than a purely centralized manual decision.

However, the implemented prototype does not claim to fully replace human judgement in real-world governance systems. AI moderation is used as a first-line filter for spam, abusive content, irrelevant reports, and unsafe images. In a production deployment, a dispute or appeal process may still be required for borderline cases.

4.6.3 Public Blockchain Approaches

While some theoretical governance models propose using public, permissionless blockchains (such as the Ethereum mainnet) to achieve maximum decentralization, these architectures are poorly suited for frequent local governance interactions. Public blockchains suffer from high transaction latency, limited throughput, and unpredictable, fluctuating gas fees that the user must pay. The proposed framework overcomes these usability challenges by utilizing a private, permissioned blockchain running a PoA consensus mechanism. This architecture provides predictable, high-performance transaction processing. Additionally, it eliminates the financial burden on citizens through the use of a backend relayer network (meta-transactions), offering a seamless, "gasless" experience.

4.6.4 Domain-Specific Blockchain Solutions

Recent literature highlights several blockchain-based complaint management systems, but these are primarily tailored for highly specific domains such as law enforcement (e.g., police complaint registries). These systems often focus narrowly on administrative efficiency, restrict access entirely to authorized personnel, and frequently lack robust mechanisms for handling large multimedia evidence. Unlike these isolated solutions, the proposed framework introduces a generalized, community-assisted approach. It actively encourages public participation through opinion polling and community upvoting. Furthermore, it efficiently manages large multimedia files (e.g., images of infrastructure damage) through decentralized off-chain storage via IPFS, seamlessly linking cryptographic hashes back to the immutable blockchain ledger.

Chapter 5

Discussion

This chapter discusses the main findings, important design decisions, current progress, and limitations of the proposed system.

5.1 Novel Contributions of the Project

The main contribution of this project is the design of a privacy-preserving citizen reporting system for local governance. The system combines blockchain, smart contracts, IPFS, AI-based moderation, and a web-based DApp into one framework. A key design decision is the separation of real citizen identity from on-chain report records. Citizens are first verified through the simulated GovID service. After verification, the system uses ticket IDs and pseudonymous key pairs for report submission. This helps citizens report public issues without directly exposing their identity on the blockchain.
Another important contribution is the use of a backend relayer. The relayer submits blockchain transactions on behalf of citizens. This improves usability because citizens do not need to manage gas fees or directly interact with blockchain nodes. The system also uses IPFS for off-chain storage. Images and larger report content are stored outside the blockchain, while only CIDs and hashes are stored on-chain. This reduces blockchain storage overhead while preserving data integrity.
AI-assisted moderation is used before reports are stored permanently. This helps reduce spam, abusive content, and irrelevant submissions. The moderation service is designed using multiple oracle workers and an aggregator, which reduces dependence on a single moderation node.

5.2 Overall System Validity

The current system design supports the main objectives of the project. The permissioned blockchain provides a tamper-evident record of report metadata. Smart

contracts provide the base for report creation, state tracking, and future government actions.
The identity mechanism supports privacy by separating real identity verification from blockchain activity. The DApp improves accessibility by giving citizens a simple interface for report submission. The relayer hides blockchain complexity from users.
The AI oracle service has been deployed as a containerized service on a VPS. This shows that off-chain moderation can be integrated with the reporting workflow. IPFS is planned to handle media and report content storage.
At this progress stage, the system is valid as a proof-of-concept. However, full validation requires complete integration between the DApp, relayer, IPFS, smart contracts, and blockchain.

5.3 Achievement of Proposed Objectives

The project has made progress toward the proposed objectives.
The first objective was to design a permissioned blockchain using PoA consensus. The architecture has been designed around a private Ethereum-compatible blockchain. This supports integrity and controlled participation.
The second objective was to use smart contracts for reporting workflows. The smart contract design includes report creation, status management, and event handling. Government-side actions are still under development.
The third objective was to develop user-friendly interfaces. The citizen-facing DApp interfaces for login and report submission have been developed. Further work is needed for government-side interfaces.
The fourth objective was to integrate AI-based moderation. A text-based AI oracle moderation service has been implemented and deployed. Image moderation is planned as a future extension.
Overall, the project has achieved good progress at subsystem level. The next important task is full end-to-end integration.

5.4 Summary of Results and Observations

Several observations were made during the current progress stage.
First, the modular architecture helped the team develop different parts of the system in parallel. Each component can be tested separately before full integration.
Second, the hybrid storage approach is necessary. Blockchain is suitable for metadata and audit trails, but not for large files. Therefore, IPFS is required for storing media and report content.

Third, the relayer is important for usability. It allows citizens to use the system without understanding blockchain transactions or gas fees.
Fourth, AI moderation should happen before IPFS upload and blockchain submission. This reduces the risk of storing harmful or irrelevant content permanently.
Finally, containerized deployment improved service isolation and deployment consistency, especially for the oracle subsystem.

5.5 Real-world Applicability

The proposed system can be useful for local governments where citizens report issues such as road damage, garbage disposal, drainage problems, broken streetlights, and flooding.
In a real-world scenario, a citizen can submit a report through the DApp. The report is verified, moderated, stored off-chain, and recorded on the blockchain. Later, the citizen can check whether the report still exists and track its status.
This can improve trust between citizens and local authorities. It can also create an auditable record of how public issues are handled.

5.6 Limitations of the Current Prototype

The current prototype has some limitations.
The identity system is simulated. It does not implement a complete real-world zero-knowledge identity system.
Although the current prototype supports both text moderation and image safety moderation, the moderation system is still limited to a controlled experimental environment. The image moderation component focuses mainly on detecting unsafe visual content. It does not fully verify whether an image is geographically authentic, whether it was captured at the reported location, or whether it was generated or manipulated using generative AI tools.
The oracle decentralization is simulated using separate containers. In a stronger real-world deployment, oracle nodes should run on separate servers or be operated by different organizations.
Full end-to-end integration is still in progress. Some components are implemented and tested separately, but the complete report submission flow must still be validated.
The system is also tested only in a controlled environment. Real-world deployment would require legal, security, and institutional approval.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project presents a blockchain-based, community-assisted, privacy-preserving reporting framework for local governance. The system is designed to address common problems in centralized reporting platforms, such as lack of transparency, weak data integrity, privacy concerns, and unreliable report quality.

The proposed framework combines a permissioned blockchain, smart contracts, a simulated identity service, a backend relayer, IPFS, AI-assisted moderation, and a web-based DApp. Each component has a clear role in the overall system.

The blockchain and smart contracts provide tamper-evident report records and workflow management. The GovID simulator and ticket-based mechanism support privacy-preserving access control. The relayer improves usability by handling blockchain transactions on behalf of citizens. IPFS supports scalable off-chain storage. The AI oracle service helps filter spam and inappropriate content before permanent recording.

At the current progress stage, several important subsystems have been designed and partially implemented. The AI moderation subsystem has also been deployed as separate oracle containers on a VPS. This shows that the proposed architecture is technically feasible as a prototype.

The main insight from this project is that blockchain alone is not enough for a practical civic reporting system. It must be combined with privacy mechanisms, off-chain storage, moderation, and user-friendly interfaces.

6.2 Future Work

The next major task is to complete full end-to-end integration. The DApp, GovID simulator, backend relayer, AI oracle service, IPFS, smart contracts, and blockchain must work together in one complete report submission flow.

The government-side workflow should also be completed. This includes report review, issue assignment, resolution updates, closure, and citizen feedback.
The current identity system should be improved in future work. A real zero-knowledge proof or self-sovereign identity mechanism can replace the simulated GovID service.
The AI moderation service can be extended with stronger multimodal models that evaluate not only image safety, but also image-report relevance, image manipulation, and location consistency. A future version may use vision-language models to compare the submitted image with the selected report category and description.
The oracle deployment can also be improved by hosting oracle workers under different administrative domains instead of running them as containers on the same VPS. This would make the oracle layer closer to a decentralized oracle network.
More testing is required after integration. This includes blockchain transaction latency, IPFS upload time, oracle response time, and complete report submission time.
Security testing is also required before any real-world deployment. Smart contracts, APIs, private keys, relayer logic, and deployment infrastructure must be audited carefully.
Finally, the DApp should be improved through usability testing. Citizens should be able to submit and track reports easily without needing blockchain knowledge.

6.3 Final Remarks

The proposed system provides a strong foundation for a transparent and privacy-preserving local governance reporting platform. Although the prototype is still under development, the current progress shows that the architecture is feasible.
With further integration, testing, and improvement, this framework can support more trustworthy citizen participation and better accountability in local governance.

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